USING AI TO AUDIT CONCURRENT ENGINEERING

A Case Study in Marine Vessel Design and VDI 2221 Compliance

Lappeenranta–Lahti University of Technology LUT

Mechanical Engineering

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Jonne Valola

Examiners: Professor Juha Varis

Associate Professor Amir Togiani, D.Sc.

ABSTRACT

Lappeenranta–Lahti University of Technology LUT

LUT School of Energy Systems

Mechanical Engineering

Jonne Valola

Using AI to Audit Concurrent Engineering: A Case Study in Marine Vessel Design and VDI 2221 Compliance

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Keywords: VDI 2221, systematic design process, Retrieval-Augmented Generation (RAG), concurrent engineering, automated project analysis, naval engineering

Marine vessel design is often described as the "art of balancing compromises." The complexity of naval engineering and challenging project timelines demand increasingly sophisticated project organization and deeply layered concurrent engineering. This research investigates whether the combination of artificial intelligence and VDI 2221 systematic design guidelines can be used to audit how a design engineering team proceeds in a project.

Two research methodologies are compared: a traditional participant observation and a fully automated process using a Retrieval-Augmented Generation (RAG) -enhanced artificial intelligence. The case study being examined is the design engineering project of a water taxi boat destined for a major European city.

The study reveals that the water taxi project is an example of a highly agile concurrent engineering process, but that the project also exhibits violations of the VDI 2221 guidelines. Most notable of these is the practice of "stage-jumping", which means, for example, advancing into detailed design before the conceptual solution is mature.

The study demonstrates that a suitably trained AI can create a high-level view of a project in a fraction of the time a human requires to do the same. It is plausible that AI could even be used for real-time project situational feedback. However, for that to be possible, a project should follow a systematic process such as VDI 2221 to provide a contextual framework for the AI analysis. In conclusion, the combination of AI and VDI 2221 could be an interesting foundation for the development of novel project management tools.

TIIVISTELMÄ

Lappeenrannan–Lahden teknillinen yliopisto LUT

LUTin energiajärjestelmien tiedekunts

Konetekniikka

Jonne Valola

Tekoäly rinnakkaissuunnittelun arvioinnissa: tapaustutkimus erään aluksen suunnittelusta verrattuna VDI 2221 -standardin ohjeistukseen

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Veneen suunnittelua kuvataan usein ”kompromissien tasapainottamisen taiteeksi.” Meritekniikan suunnitteluprojektien monimutkaisuus ja alati tiukentuvat aikataulut edellyttävät yhä korkeatasoisempaa projektien organisointia ja rinnakkaissuunnittelua (concurrent engineering). Tässä tutkimuksessa tarkastellaan, voidaanko systemaattisen suunnittelun VDI 2221 -ohjeistoa ja tekoälyä käyttää yhdessä projektiryhmän tekemän rinnakkaissuunnittelun seuraamiseen ja arviointiin.

Tutkimuksessa hyödynnetään kahta menetelmää: perinteistä osallistuvaa havainnointia sekä täysin automatisoitua prosessia, joka käyttää RAG-pohjaista (Retrieval-Augmented Generation) tekoälyä. Näitä menetelmiä sovelletaan eurooppalaiseen kaupunkiin suunniteltavan vesitaksin tapaustutkimukseen.

Tutkimuksen näkökulmasta vesitaksiprojekti on esimerkki tehokkaasta rinnakkaissuunnittelusta, mutta VDI 2221:n kannalta projektissa on poikkeamia. Projekti mm. hyppää usein yksityiskohtaiseen suunnitteluun tilanteissa, joissa tarvittava esisuunnittelu on vielä kesken.

Tutkimus osoittaa, että tekoälyllä voidaan luoda nopeasti yleiskatsaus projektiin. Jopa reaaliaikainen projektin tilan seuranta voisi olla mahdollista. Tämä edellyttäisi, että projektissa noudatettaisiin systemaattista prosessia, kuten VDI 2221 -standardia, jotta tekoälyanalyysille olisi kontekstuaalinen viitekehys. Tekoälyn ja VDI 2221:n yhdistelmä voisikin toimia mielenkiintoisena perustana täysin uudenkaltaisten projektinhallintatyökalujen kehittämiselle.

ABBREVIATIONS

AC	Axial Coding
AI	Artificial Intelligence
BOM	Bill of Materials
CAD	Computer-Aided Design
CE	Concurrent Engineering
FEA	Finite Element Analysis
GenAI	Generative Artificial Intelligence
GT	Grounded Theory
LLM	Large Language Model
MBSE	Model-Based Systems Engineering
OC	Open Coding
PBD	Point-Based Design
RAG	Retrieval-Augmented Generation
SBD	Set-Based Design
TDM	Text and Data Mining
VDI	Verein Deutscher Ingenieure (Association of German Engineers)

GLOSSARY

Aft	Synonym for rear.
Bow	The forward-most part of a vessel's hull. It is the first point of contact between the ship and the water, and its design is critical to the ship's performance, efficiency, and stability.
Bulkhead	Vertical partition within the hull of a ship. By dividing the hull into watertight sections, bulkheads prevent a single breach from sinking the entire ship.
Chine	Longitudinal intersection where the bottom of a ship's hull meets the sides. It often represents a distinct change in the angle of the hull's cross-section but may also be completely round and smooth.
Fitment	The installation and integration of non-structural components onto a ship's hull.
General Arrangement	Architectural plan that defines how the ship is divided into functional spaces and where every major piece of equipment is located.
Scantling	The collective dimensions, materials, and thicknesses of a ship's structural members.
Stern	The aft-most (rear) part of a ship's hull. The stern determines the wake of the vessel and needs to support rudder and propulsion forces.
Transom	The “rear wall” of the vessel, usually flat or slightly curved. Important in the structural strength of the vessel.

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DECLARATIONS

Turnitin

The originality of this thesis has been reviewed with the Turnitin similarity checking service.

AI usage

The author of the thesis, Jonne Valola, used the following AI-tools during the preparation of the thesis:

1. Miro AI
    a. Converting mindmaps into categorized lists.
    b. Miro is an on-line tool, which allows for the consolidation of different types of source material (images, texts) onto a virtual digital whiteboard. Miro AI can be prompted to convert visual graphs into categorized text automatically. These categorized lists were not used as-is in the thesis work, but as a tool for considering alternatives when preparing the manually written final lists.
2. Dr. Marine (Gemini 3 pro – based text and data mining agent created by author)
    a. Generating analysis of project documents and observations.
    b. Used as an advanced find and filter tool for tracking the evolution of the design of the vessel from project documentation and observations. These analyses are part of the research material.
3. Microsoft Copilot
    a. Generating graphs from excel files.
    b. Used to convert data of coded observations into graphs of the frequency of discussion of different topics.

4. Google Gemini 3 Pro
    a. Searching for boat market data and other resource data.
    b. Used to text and data mine for information which was difficult to find using traditional web searches.
5. Sembly AI
    a. Recording and summarizing meetings.
    b. Used in meetings to record, transcribe and automatically generate meeting notes. Used only as input data to observations and analysis.
6. Nano Banana Pro
    a. AI image and diagram creation tool.
    b. Used for diagrams.

Responsibility

The author, Jonne Valola, takes full responsibility for the content of this thesis and has reviewed and edited the content generated by the possible use of AI tools.

1 Introduction

Modern watercraft are sophisticated “systems of systems”. For example, propulsion, steering, cargo loading and unloading, passenger facilities, navigation instruments and safety equipment all need to fit on board and function in unison to create a safe and effective marine vessel. The classic axiom of ship design states that naval architecture is the “art of making a compromise between conflicting requirements.” In the introduction to their seminal book “Basic Ship Theory” (2002, 1) naval architects and authors K.J. Rawson and E.C. Tupper say of the requirements for ship safety, economy and performance that, “It must be clear that naval architecture involves complex compromises of many of these features. The art is, perhaps, the blending in the right proportions. There can be few other pursuits which draw on such a variety of sciences to blend them into an acceptable whole. There can be few pursuits as fascinating.” In “The Nature of Ship Design” (1998, 187) Andrews states that all systems essentially compete for the same space and weight budget of the vessel: “The ship design task is essentially one of integrating a wide range of conflicting requirements -- principally the accommodation of payload and equipment within a hull of limited dimensions and displacement, where the demand for internal volume and the restriction of total weight are often in direct conflict.”

This study used ethnographic methods, such as participant observation and AI-driven automation to track the design delivery process of a vessel at a marine design agency. The tracking was done by following project design meetings and the evolution of the design documents in minute detail. At the onset of the research, the case study “Watertaxi” project was at an ideal stage for observation, as the kick-off meeting of the project had been only a few weeks before. Permission to join in to the meetings was received from the project team and the client. Recordings and logs of past meetings allowed mapping the entire history of the Watertaxi project to the point where the study began.

1.1 Concurrent Engineering in naval design

In engineering, research and development of the different aspects of a product or service is carried out both sequentially and in parallel. An engineering philosophy called “Concurrent

Engineering" (CE) has become the popularized synonym for highly organized simultaneous engineering. The aim of CE is to parallelise design and engineering work to cut development time and to minimize late-stage engineering changes. The term Concurrent Engineering was originally coined by Winner et al., in a report published in 1988 by the Institute for Defense Analyses (IDA). In the report, Concurrent Engineering is described as "a systematic approach to the integrated, concurrent design of products and their related processes, including manufacture and support. This approach is intended to cause the developers, from the outset, to consider all elements of the product life cycle from conception through disposal, including quality, cost, schedule, and user requirements." One of the early advocates of CE, Dr. Biren Prasad, distinguishes parallel scheduling as a distinctive feature. "Concurrent Engineering is a work methodology based on the parallelization of tasks (i.e. performing tasks at the same time), which is sometimes called simultaneous engineering." (Prasad 1996, 1).

It is important to make the distinction, that CE is both a general term used to describe parallelisation of engineering and a marketing used by experts who provide services to management. The usage of CE in the case of this study refers to the general idea of coordinating several areas of engineering work in parallel. In this study, CE does not refer a particular style of concurrent engineering and does not attempt to critique the benefits and disadvantages of CE itself.

The ideation and development of a marine vessel require designing many aspects of the craft at the same time. The various systems also compete for physical space on board the vessel. The list of solutions needed for the Watertaxi gives an illustration of the breadth of engineering required even for a relatively small marine vessel (Table 1). In the Watertaxi project, the project schedule was extremely parallelised. The Watertaxi team was aiming to design propulsion systems, steering systems, hull design, passenger access and comfort, driver ergonomics, thermal management etc. within the same timeframe. While there is nothing unusual about such a situation in engineering, the Watertaxi was notable for the lack of having even one subsystem that would have been fully mature before the development project began. In interviews with project team members (Appendices 2 and 3) it became evident, that the Watertaxi was unusually "open" with regards to the definition of the engineering of different systems at the onset of the project.

Table 1. Systems developed for the Watertaxi.

System	System components
Propulsion	Electrical powertrain consisting of high voltage batteries, cabling, inverters, motor controllers, two motors, motor beds, drive shafts, thrust bearings and propellers, bow thruster
Steering	Two custom-fabricated rudder plates, rudder stock, bearings, rudder lever arms, electro-hydraulic actuators, steer-by-wire driving wheel and trimming tabs
Hull	Marine-grade all-aluminium welded hull optimized for light weight with reinforced frames, longitudinals and superstructures able to withstand impacts of constant docking maneuvers and category C wave conditions
Cabin	Weather-protected seating for 12 passengers with windows, pneumatically assisted door, polycarbonate windows, aft deck with outdoor benches, access hatches to powertrain systems underneath, emergency escape corridor and life vests
Thermal systems	Powertrain cooling systems, battery thermal management systems, battery de-gassing ventilation fans and ducts, cooling system pumps and ducting, fireproofing insulation, gas and thermal runoff detection systems, automatic extinguishers
Driver systems	Driver seat pedestal with suspension, boat systems control interface with display, communication, radar, GPS navigation and steering controls, taxi route management and passenger charging system, emergency equipment
Electrical systems	High-voltage system, low voltage circuit, fuses, battery quick disconnect couplings, bilge pumps, cooling pumps, battery chargers, DC-DC converters, lighting and signaling

1.2 Research objective

This study investigates the concurrent engineering processes of the Watertaxi case study by evaluating real-world processes against the framework of VDI 2221. The evaluation is carried out by following design meetings and design documents in minute detail. This is done by both traditional participant observation and a fully automated AI-analysis. The objective of the research is to discover whether this approach is able to provide high-level project analysis. The ultimate goal would be to use this kind of methodology to improve and facilitate the management of a complex concurrent engineering project.

1.3 Research questions

The study focuses on the following research questions:

Question 1: “Can artificial intelligence be used to analyze and assist in the complicated process of marine vessel design?”

Question 2: “What specific patterns of non-adherence to VDI 2221, if any, does the AI detect in a real-world naval engineering project?”

Question 3: “How do these AI-detected deviations correlate with results from a traditional participant observation done by a human?”

Question 4: “What is the potential of AI tools to assist in real-time monitoring of the trajectory of a project?”

1.4 Scope and limitations

Coming from the outside, learning the working process of a naval engineering team is a daunting task. This study followed only a single project over a period of only 3 months and 21 project meetings. Even such a limited scope of observations yielded 770 pages of meeting transcripts, dozens of project documents and 4 interviews (of which 2 were used). Several case studies and longer periods of observation would of course be beneficial to the accuracy of research such as this, but time and resources were not available to extend the study further.

Qualitative analysis methods rather than quantitative methods were chosen for the research because the subject of the observation was the flowing human interaction of a project team. Forming credible advance hypotheses of what would be expected to happen within the project would have been equally challenging. Therefore, the gradual evolution of research hypotheses enabled by the methodology of Grounded Theory was chosen as a means of digging deeper into the research subject.

In the course of the study, it became apparent that much deeper, multiple prompt analysis of the project data would also be possible. The benefits and results of those investigations is however beyond the scope of the work and are only provided as examples (4.5).

2 Theory

For collecting data, this study used methods established by Grounded Theory as a foundation for participant observations and coding of the observations. For evaluating and structuring the data, the study used the framework of VDI 2221 Systematic Engineering Design process as a "yardstick of maturity" for different design threads. The final analysis of the study used a Retrieval-Augmented Generative AI to automate the analysis of the observations. The theoretical backgrounds of these concepts are explained here.

2.1 Grounded Theory

The term "Ethnography" derives from the Greek language, combining "ethnos" (ἔθνος) meaning "people or cultural group" and "grapho" (γράφω) meaning "to write." Hammersley and Atkinson (2007, 3) describe ethnography as a research activity, in which the researcher participates in the phenomenon being studied. From this vantage point, embedded in the subject of research, the researcher observes, listens and records people's daily activities, collecting data which may shed light on the broader issues.

An ethnographic researcher begins from a distance to the subject. It must be considered how to get closer the the subject and gain understanding of it. The classical approach to research is to first form a hypothesis, i.e. an educated guess about the research subject. The researcher would begin by studying existing literature, formulating an approximate theory, and then designing experiments to prove or disprove the hypothesis. In Grounded Theory (GT) the researcher starts with as few assumptions as possible. The researcher accepts not knowing and not guessing why a problem or a phenomenon exists. Theory building process begins with pure observation, and the emerging data will inform the researcher what should be studied in more detail as the study progresses. Grounded Theory is a systematic methodology used to induce a theory from data, rather than starting with a hypothesis and testing it. GT methods provide steps to analyze field notes, code the data, find patterns and begin forming theory of the phenomena.

> "Grounded theory methods consist of systematic, yet flexible guidelines for collecting and analysing qualitative data to construct theories from the data themselves." (Charmaz 2014, 1).

2.1.1 Open Coding (OC) and Axial Coding (AC)

In Grounded Theory, Open Coding (OC) and Axial Coding (AC) are the two sequential stages of analyzing qualitative data.

Open Coding is the first step, where the researcher goes through the data and labels fragments of it with identification codes. These codes can be simple key words such as "assumption" to signify that a certain remark was made without further verification.

In the second step, Axial Coding, the aim is to find relationships between the coded fragments. There may be several relationships, for example "requirements", meaning the observation refers to project requirements, and "propulsion", meaning the observation also belongs to the propulsion subsystem.

2.2 The VDI 2221 Systematic Design process

As the modern industrial age progressed, manufacturing systems evolved and required product functionality became increasingly complex. The Association of German Engineers (Verein Deutscher Ingenieure - VDI) initiated in 1970's a search for an alternative model to the traditional, intuition- and experience-based product research and development processes. This led to the first formalizations of "Systematic Engineering Design" by industry pioneers, professors Gerhard Pahl and Wolfgang Beits, and the publication of the VDI 2221 guideline "Methodik zum Entwickeln und Konstruieren technischer Systeme und Produkte" – "Systematic Approach to the Design of Technical Systems and Products". The VDI 2221:1993 guideline has become an unofficial standard for systematic thinking as practised by engineers educated in Germany and is recognized throughout the world.

In their book, "Engineering Design: A Systematic Approach" (2013, 1) Pahl & Beitz introduce the aim of the VDI guideline 2221: "(it) defines an approach and individual methods for the conceptual design of technical products and is therefore particularly suitable for the development of new products -- (it) proposes a generic approach to the design of

technical systems and products, emphasising the general applicability of the approach in the fields of mechanical, precision, control, software and process engineering. The approach (Figure 1) includes seven basic working steps that accord with the fundamentals of technical systems and company strategy. Both guidelines have been developed by a VDI Committee comprising senior designers from industry and many of the previously mentioned design methodologists from the former West Germany."

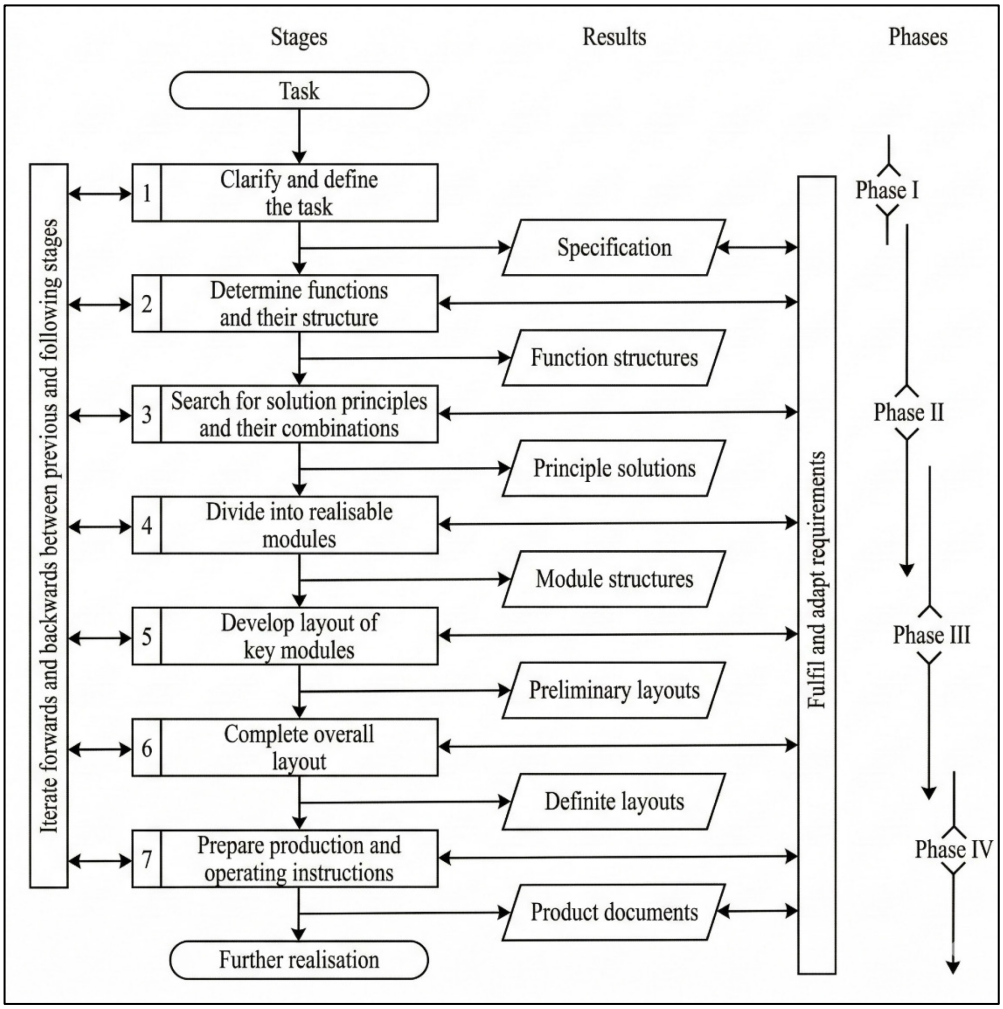


Figure 1. General approach to systematic design (based on Pahl & Beitz 2013, 19).

The steps proposed by VDI 2221 are ideal for the categorization of simultaneous engineering tasks. "The division of the design process into phases and steps -- makes the process transparent and manageable. It facilitates the division of labour -- and allows for a timely parallelization of activities (Simultaneous Engineering)." (Pahl & Beitz 2013, 17). The seven steps of VDI 2221 process are summarized and paraphrased by the author in Table 2.

Table 2. The seven steps of systematic design (based on VDI 2221:1993).

Step	Original German Term	English Translation	Key Characteristics & Output
1	Klären und Präzisieren der Aufgabenstellung.	Clarify and define the task.	Understanding what is needed, not how to build it. Output: requirements list or functional specification.
2	Ermitteln von Funktionen und deren Strukturen.	Determine functions and their structures.	Abstracting the problem without specifying exact technical solutions. Output: abstraction of the functional module as a “Black Box” of inputs and outputs.
3	Suchen von Lösungsprinzipien und deren Strukturen.	Search for solution principles and their combinations.	Brainstorming how to achieve the functions on a conceptual level, e.g. by comparing solutions based on different principles of physics or available technologies. Output: conceptual solution.
4	Gliedern in realisierbare Module.	Divide into realizable modules.	Breaking the complete expected system into subsystems and defining the interfaces between them. Output: schematic of subsystems as modules and their interaction.
5	Gestalten der maßgebenden Module.	Develop key modules.	Embodiment Design, i.e. selection and definition of functionally critical parts and their physical layout. Output: preliminary layout.
6	Gestalten des gesamten Produktes.	Develop entire product.	Detailed design. Exact specification of components and dimensions. Adding auxiliary systems and refining final layout. Output: design freeze.
7	Ausarbeiten der Ausführungs- und Nutzungsangaben.	Prepare production and operating instructions.	Documentation for manufacturing, including CAD models, technical drawings, bills of materials (BOM) and manuals. Output: product and manufacturing documentation.

2.3 Generative Artificial Intelligence (GenAI)

Research into the application of Artificial Intelligence to different practical tasks has been a growing topic in academic studies. “Harnessing Artificial Intelligence for Project Management Efficiency” by Oscar Tyrberg and Tim Adenmark (2024) provides an excellent overview of the current state-of-the-art, especially in the context of using AI tools for project management. “Usage of Artificial Intelligence in Industrial Design Processes” by Esa Tammisto (2025) offers another broad look into the opportunities of using AI within the design process. However, the landscape of AI is evolving so rapidly, that even these new studies have no knowledge of things have changed since with newer and more powerful AI offerings, for example Google’s Gemini 3 model released in November 2025.

Generative Artificial Intelligence (GenAI) refers to AI models which are able to create human-like responses to queries. GenAI models have been trained with textual, visual and audio data from vast existing datasets. GenAI is able to recombine the patterns and styles it has been trained on to create new, different output. Its’ primary use is in art creation, music composition and text generation. Popular GenAI tools in 2026 include Midjourney, DALL-E 3 and Stable Diffusion for image creation, Sora and Runway for video creation and Suno for music. Text-output focused GenAI’s are a group called Large Language Models (LLMs).

2.3.1 Large Language Models (LLMs)

Large Language Models (LLMs) are a subset of GenAI focused on human language processing. According to Zhao et al. (2023), a LLM is a deep learning algorithm trained on massive datasets of text. Fundamentally, it functions as a probabilistic engine: it analyses the statistical relationships between words (tokens) to predict the most likely next token in a sequence, allowing it to generate coherent, human-like text and perform complex reasoning tasks (Figure 2).

2.3.2 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is a hybrid AI framework. In RAG, the accuracy and reliability of LLM reasoning is improved by providing the AI model with access to

additional external data, which was not originally included in the training of the foundational model of the neural network. In answer to a user prompt, a RAG system first looks at the additional, external data and then uses relevant portions of that data with the foundational LLM engine to formulate a reply (Figure 3). A RAG is also able to argument and refer to specific information in the external data. This makes it more reliable in situations, where for example, a word-accurate requirement set out by a standard is sought for.

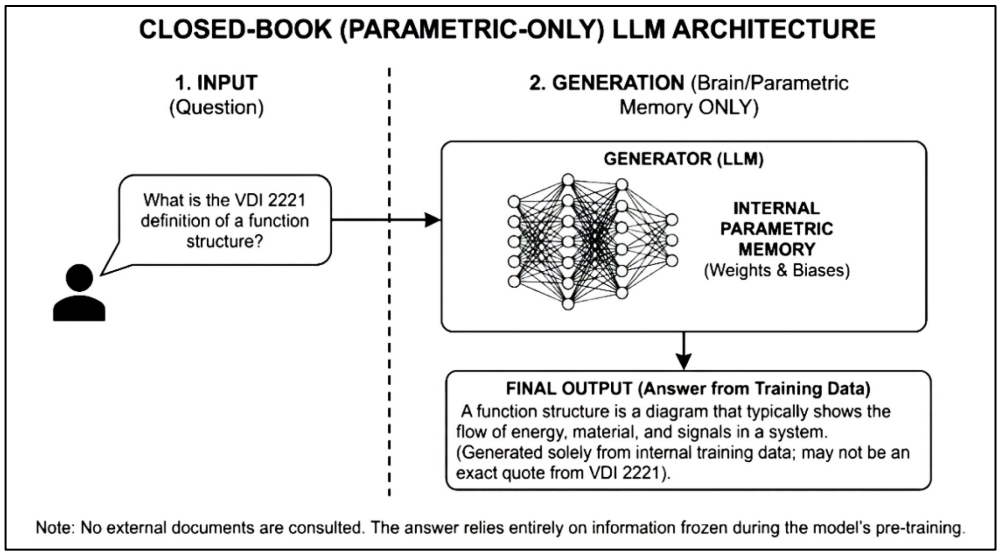


Figure 2. Response system of a closed-book LLM (Generated with Nano Banana Pro).

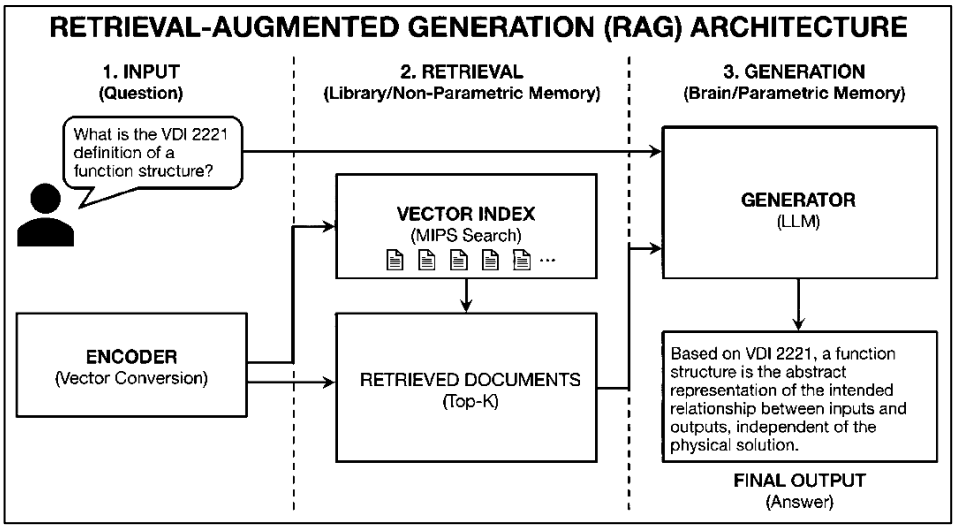


Figure 3. Response system of a Retrieval-Augmented Generative AI (Generated with Nano Banana Pro).

2.3.3 Text and Data Mining (TDM)

Text and Data Mining (TDM) is the automated process of selecting and analysing large amounts of text or data resources for the purposes of searching for patterns, trends, and correlations. It involves using computational techniques to extract meaningful information and knowledge from unstructured (text) and structured data sets.

Text and Data Mining (TDM) refer to using automated tools, such as LLMs, to seek through, analyze and collate large amounts of interconnected data. In contrast to traditional data analysis, LLMs are able to use reasoning to find relationships between dissimilar and differently formatted data. In TDM, one source of data can also be used as a "filter" to find and create analysis of another data from a very specific point of view.

2.3.4 Legal considerations of using TDM for scientific research

This study used Text and Data Mining to analyze project documentation and recorded observations. This included using copyrighted literary materials as input to analyze other sets of data. European Union's "Directive on Copyright in the Digital Single Market", officially known as Directive (EU) 2019/790, specifically provides Article 3, which permits using lawfully accessed copyrighted works in Text and Data Mining for the purposes of scientific research (European Parliament and Council of the European Union, 2019, Art. 3).

3 Methods

Project tracking for the “Watertaxi” was done for the time span of 19.8.2025 to 25.11.2025. The data was evaluated and coded into observations by two methods: manually and by using a fully automated and enhanced AI tool. The enhanced AI tool was a Google Gemini 3 Pro-powered Large-Language Model (LLM) chatbot which had been improved with RAG (Retrieval-Augmented Generation) by adding topic-specific external knowledge to its reasoning model. The coded observations were evaluated and structured using the theoretical framework of the VDI 2221 guideline “Methodik zum Entwickeln und Konstruieren technischer Systeme und Produkte” – “Systematic Approach to the Design of Technical Systems and Products”. The structured data was then further processed to reveal patterns and details about the workflow of the project, highlighting recurring sources of problems and areas of potential improvement. Finally, the enhanced AI tool was asked to review the entire project flow based on the observations and to give recommendations on how to improve the design delivery process.

3.1 Data collection

3.1.1 Participant observation (the manual process)

Table 3 lists the 21 meetings which were observed and coded. The meetings consisted approximately 42 hours of discussion.

Table 3. Participant Observations.

Meeting date	ID	Topic	Participation type
19.8.2025	M1	Weekly meeting	Recording
9.9.2025	M2	Weekly meeting with the whole team	Recording
16.9.2025	M3	Weekly meeting with the whole team	Recording
22.9.2025	M4	Nesting details	Recording

23.9.2025	M5	Weekly meeting with the whole team	Recording
26.9.2025	M6	Fitment & weight meeting	Recording
30.9.2025	M7	Weekly meeting with the whole team	Recording
1.10.2025	M8	Weight meeting Agency & Client	Recording
7.10.2025	M9	Weekly meeting with whole team	Recording
14.10.2025	M10	Client visit at Agency – weight & fitment	Live
16.10.2025	M11	Weekly meeting with whole team	Recording
21.10.2025	M12	Weekly meeting with whole team	Live
4.11.2025	M13	Weekly meeting with whole team	Live
6.11.2025	M14	Cabin & interior review meeting	Recording
6.11.2025	M15	Powertrain & Cooling meeting	Recording
11.11.2025	M16	Weekly meeting with whole team	Live
13.11.2025	M17	Concept details and cooling	Recording
13.11.2025	M18	Cooling and rudder layout	Recording
18.11.2025	M19	Weekly meeting with whole team	Live
18.11.2025	M20	Weight status meeting	Live
25.11.2025	M21	Whole team meeting	Recording

3.1.2 Project document analysis

The Watertaxi project documentation was made available remotely via a Microsoft Teams channel. This channel was used by all parties, including the clients and collaborating external teams. This gave a very comprehensive view into how the project evolved and how the project team proceeded. The Watertaxi document analysis focused mainly on the overall project status report. The Watertaxi project teams used this project status report as a “single-source-of-truth”, meaning all important meeting notes, decisions, and email discussions were collected into it. By 25.11.2025 the Watertaxi project document consisted of 155 pages. The Watertaxi documents were used mainly for the purpose of cross referencing and verifying the data gained from the meeting transcripts.

Table 4. Documents analysed.

Document ID	Type of document	Timespan covered by document
WT_D1	Overall status report Watertaxi project. Collation of meeting notes and important emails, including images and other data.	12.8.2025-25.11.2025
WT_D2	Responsibility matrix of Watertaxi project. Describes division of tasks and responsibilities for crosschecking and approving output of other teams.	12.8.2025-25.11.2025
WT_D3	Initial schedule from Watertaxi Client side.	16.9.2025
WT_D4	Initial parts list and their locations from Watertaxi Client side.	25.9.2025
WT_D5	Safety requirements for small ships. Initial requirements provided by Watertaxi Client.	9.9.2025
WT_D6	Initial construction design principles and specification from Watertaxi client.	2.10.2025
WT_D7	Evolving building specification from Watertaxi Client.	19.11.2025-11.12.2025
WT_D8	Master parts list from Watertaxi Client.	19.11.2025-11.12.2025
WT_D9	Minutes of meeting - Kick-off of Watertaxi project.	1.7.2025
WT_10	Minutes of meeting – first weekly meeting of Watertaxi project.	9.9.2025

3.1.3 Anonymizing of meeting logs and project documents

Before analysing documents with automated AI tools, some sensitive details, such as names of participants from the Agency, the Client and Externals were removed. In addition, images were removed from the documents. The anonymization process was done by means of a Word Macro, which automatically replaced all identifiable names with single first initials and removed all images from the document. Component supplier names were left intact. This was a conscious choice as the suppliers of boat system components are well known within the industry.

3.2 Data structuring by manual coding

Manual coding of observations was done in accordance with the steps described by Grounded Theory.

3.2.1 Open Coding

Processing the data began with Open Coding, meaning that each manually logged observation was labelled with a keyword which described the approximate reason why each particular discussion or detail had attracted attention. The coding began simply by stopping at each observation, considering what could be a descriptive word and then tagging the observation with that word. After coding approximately 100 observations, it became clear that many of the code words were synonyms for the same underlying issue, such as “missed information” and “miscommunication” - both being communication problems. After consolidating similar codes and meanings, a schema of repeating “codes” began to emerge quite naturally. After processing more observations and several similar consolidation stages later, the codes used had finally settled to the 12 listed in Table 5.

Table 5. Keys used in Open Coding.

Code	Meaning
Concurrency	(issue) Task A and Task B needed to happen simultaneously, but there have not been enough resources to do so.
Dependency	(issue) Task A cannot move forward because waiting for Task B.
Chicken and egg	(issue) Task A is waiting for Task B, Task B is waiting for Task A.
Communication	(issue) Difficulties in transmitting information, missed information etc.
Assumptions	Assuming something without clear basis. May lead to false positive situation assessment.
Concept	(issue) The solution to Task A is undefined i.e., no technical or other data to describe it.
Postponement	(issue) Decision is “kicked down the road,” usually due to lack of input data. May lead to lapses in task tracking.
Requirements	(issue) The true requirements are not known.

Human factors	Operator/passenger/maintenance behaviour factor.
Decision	Task closed successfully.
Allocation	A task assigned to person / party.
Accuracy	Task delayed because accuracy of deliverable from previous step was not accurate.

3.2.2 Axial Coding

After Open Coding, Axial Coding was used to link the individual observations into related items. In Grounded Theory, the axial codes are usually invented by the researcher based on analysis of the relationships and groups of the open coded observations. On examination of the open coded observations in the case of the Watertaxi, it was evident that no invention of Axial Coding groups was needed. The observations could quite naturally be grouped according to the VDI 2221 maturity of the design thread being discussed (Table 6), and also by the vessel subsystem (Table 7). Experimenting with this Axial Coding system seemed to produce coherent results. Project management and weight discussion -categories were added as individual groups in addition to actual subsystems, because so much of the discussion was dedicated to these two topics.

Table 6. Keys used in coding of VDI 2221-phases.

Code	Meaning
Clarification	Item belongs to the requirements clarification phase of the VDI 2221 process.
Concepting	Item belongs to the concept design phase of the VDI 2221 process.
Embodiment	Item belongs to the embodiment design phase of the VDI 2221 process.
Detail	Item belongs to the detailed design phase of the VDI 2221 process.
Implementation	Item belongs to the implementation phase of the VDI 2221 process.
Verification	Item belongs to the verification phase of the VDI 2221 process.

Table 7. Keys used in coding of observations according to functional subsystem.

Code	Meaning
Project management	Item concerned the organization, communication or other interaction of teams within the project.
Exterior	Item concerned the exterior design of the boat, including interfacing with the dock and charging systems.
Weight	Item was directly concerned with discussion of weight of the vessel. Separated as its own category due to its prominence.
Hull	Item directly concerned the welded aluminium structure of the hull.
General Arrangement	Item concerned the general layout i.e. bulkheads, battery and engine compartments, seating which affect the hull structure
Fitment	Item belongs to fitment i.e. detailed placement and types of non-load bearing equipment fitted into the structure, such as motors, driveshafts, navigational controls, seats and emergency equipment.
Cabin ergonomics	Item concerned human factors and the internal design of the cabin.
Propulsion	Item concerned the means of moving the boat, including bow thruster.
Rudder & steering	Item belongs to the steering subsystem, including rudder implementation.
Batteries & battery cooling	Item belongs to the battery and battery cooling subsystem, including cooling plates, battery management systems and battery ventilation.

3.3 Data structuring by automatic coding

3.3.1 Rationale behind using a Retrieval-Augmented AI for analysis

The decision to create a custom Retrieval-Augmented AI agent came from examining the meeting notes which Sembly AI automatically created from meeting transcripts. By carefully comparing the meeting notes with the actual recorded audio and the transcript of the meeting, it was apparent that the lack of naval design -specific knowledge was crippling the reasoning of the LLM. For example, frequent references to “scantling” by the meeting participants were recorded as “scanning” in the meeting transcripts by Sembly. While a reader with knowledge of the context will understand that the recording refers to scantling, these types of errors raised suspicion about the accuracy of the interpretation done by Sembly.

3.3.2 Dr. Marine, an AI with expertise in VDI 2221 and naval engineering

Picaxe.co, an online platform, was used to create Dr. Marine, an AI chatbot with naval engineering and systematic design expertise. Picaxe.co allows users to choose the LLM reasoning model and add knowledge to enable Retrieval-Augmented Generation. The chatbot was tested with three different foundation LLM's: ChatGPT 4.1, ChatGPT 5 and Google Gemini 3 Pro. Side-by-side comparison of the same project meeting transcripts indicated, that Gemini 3 Pro provided the most accurate interpretations of the discussions in the meeting transcripts. Input length tokens were increased to 228705 tokens to allow the processing of longer document sections. Output length tokens were kept at a modest 7213 as only concise, short replies were required. Randomization was kept at default 0.8 value. The material uploaded as RAG knowledge data included copyrighted literary materials. These materials were used in accordance with European Union's Directive 2019/790, Article 3, which permits using lawfully accessed copyrighted works in Text and Data Mining for the purposes of scientific research. In compliance with Article 3, all input material used for Text and Data Mining in this study were legally obtained and are listed in Appendix 4.

For the user interface of the chatbot, a friendly icon of an expert wearing both a doctor's stethoscope and a captain's hat was generated inside the Picaxe.co editor. Finally, the prompt text seen in Figure 4 was given as a guidance to the chatbot to behave as an expert in naval engineering and systematic design.

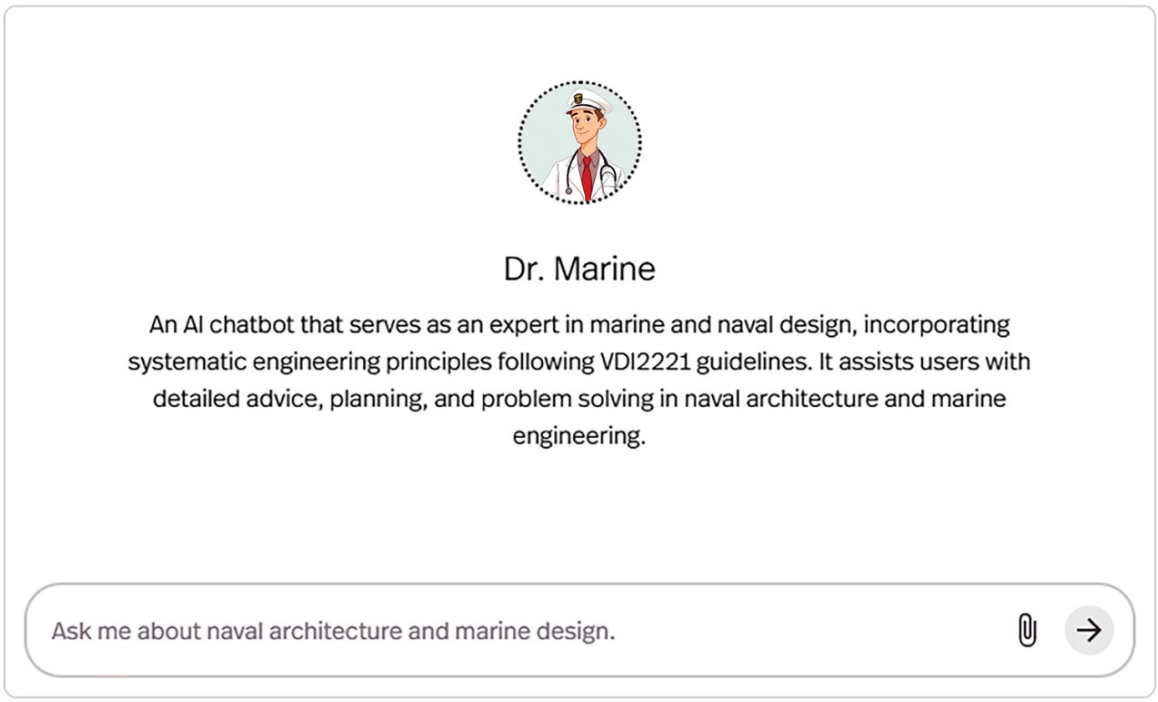


Figure 4. User prompt window to Dr Marine AI.

3.3.3 The automatic data coding process

In contrast to the manual coding, which took several days, the automatic coding process of a total of 771 pages of meeting transcripts took less than 2 hours. In the automatic coding process, meeting transcripts were input to Dr. Marine, one meeting at a time, and the following prompt was used:

> "Make a chronological list of all the different topics that are discussed in this meeting transcript. For each item, add an estimate of to which stage of the VDI 2221 systematic design process the item belongs to (clarification, concept, embodiment, detail, verification). For each item, add the identification of the subsystem, to which the item belongs. The categories are "fitment", "hull", "batteries & battery cooling", "steering", "exterior design", "weight discussion", "propulsion", "project management & task allocation", "rudder & steering", "cabin ergonomics". If you are unable to process the whole document, give a warning."

A sample of the output given by Dr. Marine when input this prompt and a meeting transcript of 18.11.2025 is shown in Figure 5.

Chronological Meeting Topic List

Time	Topic	VDI 2221 Stage	Subsystem
00:30	**Agenda Setting & Status Check** Opening the meeting and defining the action points to be discussed.	**Clarification** (Clarification of the task)	**Project Management & Task Allocation**
01:07	**Shaft & Rudder Arrangement** Discussing the integration of the shaft and rudder, and the status of drawings from "Blockland."	**Embodiment** (Definitive layout)	**Propulsion / Rudder & Steering**
03:10	**Rudder Lever Arm & System Fit** Checking if the proposed rudder system fits based on the lever arm length provided.	**Verification** (Checking against requirements)	**Rudder & Steering**
04:05	**Seat Spacing** Confirming the spacing between seats and the path width.	**Detail** (Finalizing dimensions)	**Cabin Ergonomics**
04:51	**Cooling Plates Check** Verification that cooling plates have been checked in the model.	**Verification** (Design review)	**Batteries & Battery Cooling**
04:56	**Enclosed Driver Area** Discussing the sketch and pending weight calculation for the driver area.	**Embodiment** (Preliminary form design)	**Exterior Design**
05:14	**Life Vest Selection** Debating the type (aviation vs. standard), weight (2kg vs lighter), and certification guidelines (ES-TRIN).	**Concept** (Selecting solution variants)	**Fitment**

Figure 5. Sample output of a list of observations by Dr. Marine AI agent.

The output from Dr. Marine was copied into a text document, which was converted into a spreadsheet table in Microsoft Copilot with the following prompt:

> "Turn this list of discussion topics into an excel table with 4 columns. Column 1 is 'Number', Column 2 is 'Topic', Column 3 is 'VDI 2221 stage', Column 4 is 'Subsystem'. Include all text of topic, do not shorten the description."

A sample of the output given by Copilot to this prompt and the automatically coded observations from the meeting of 18.11.2025 is shown in Figure 6.

1	Topic Description	VDI 2221 Stage	Subsystem
430	Agenda & Action Points – Reviewing the agenda and previous action points regarding shaft/rudder arrangements.	Clarification	Project Management Allocation
431	Shaft & Rudder Status – Update on the shaft, rudder, and propeller arrangement and coordination with supplier (Blockplant).	Embodiment	Propulsion
432	Rudder System Fitment – Checking if the proposed rudder system fits based on the lever arm length provided.	Verification	Rudder & Steering
433	Seat & Path Spacing – Confirming the spacing between seats and the center path width.	Detail	Cabin Ergonomics
434	Cooling Plates Check – Confirmation that cooling plates have been checked in the model.	Verification	Batteries & Battery Co
435	Enclosed Driver Area – Discussion on the sketch and pending weight calculation for the driver area.	Embodiment	Exterior Design
436	Life Vest Selection – Debating between aviation vs. standard vests, weight implications, and ES-TRIN regulations.	Concept	Fitment

Figure 6. Sample output of conversion using Microsoft Copilot.

3.4 Conversion of coded data into visual bubble graphs

The datasets of both manually coded data (418 observations) and automatically coded data (505 observations) were turned into bubble graphs visualizing the frequency of appearance of each topic within the meeting in each of the 21 meetings from 19.8.2025 to 25.11.2025. This was done in Microsoft Copilot with the following prompt:

> "Make a bubble graph for this excel document. X-axis is dates (column D), Y-axis is VDI 2221 stage (column B). Size of bubble is the number of instances a subsystem (Column C) is mentioned on a given date. Bubbles should not be directly on top of each other but staggered slightly. Ignore entries where either data in column C is empty or data in column D is empty. Scale bubbles 2x. Do one graph for each subsystem and a combined graph overlaying all subsystems."

4 Results

The frequency of different topics within the Watertaxi project meetings was mapped both manually and using AI automation. In both cases, an evaluation was to be made as to what VDI 2221 design stage each observation belonged to and what topic it concerned.

4.1 Bubble graphs as a graphical representation of discussion activity

The bubble graphs in the following section are a graphical representation of what the Watertaxi project team discussed in their design meetings. In each topic, both a manually coded data set and an AI-automated data set is shown. The horizontal represents the time of observation. The vertical axis is the detected VDI 2221 systematic design phase. The size of the bubbles indicates the number of times a subsystem topic emerged within a meeting. Figure 7 shows the manual mapping of all discussion topics over 21 meetings (418 datapoints). Figure 8 shows the same for the AI automated analysis (505 datapoints).

The purpose of representing the data as bubble charts in this way is to visualize how the discussion of the design of different subsystems evolved over time. Side-by-side comparison of the “manual” and “automatic AI” bubble graphs also shows any differences in the way the results were interpreted in a manual process (participant observation) and an AI-automated process. As it can be seen, the distribution of topics, and especially the distribution of the division of observations, is at times significantly different between manual and automatic coding.

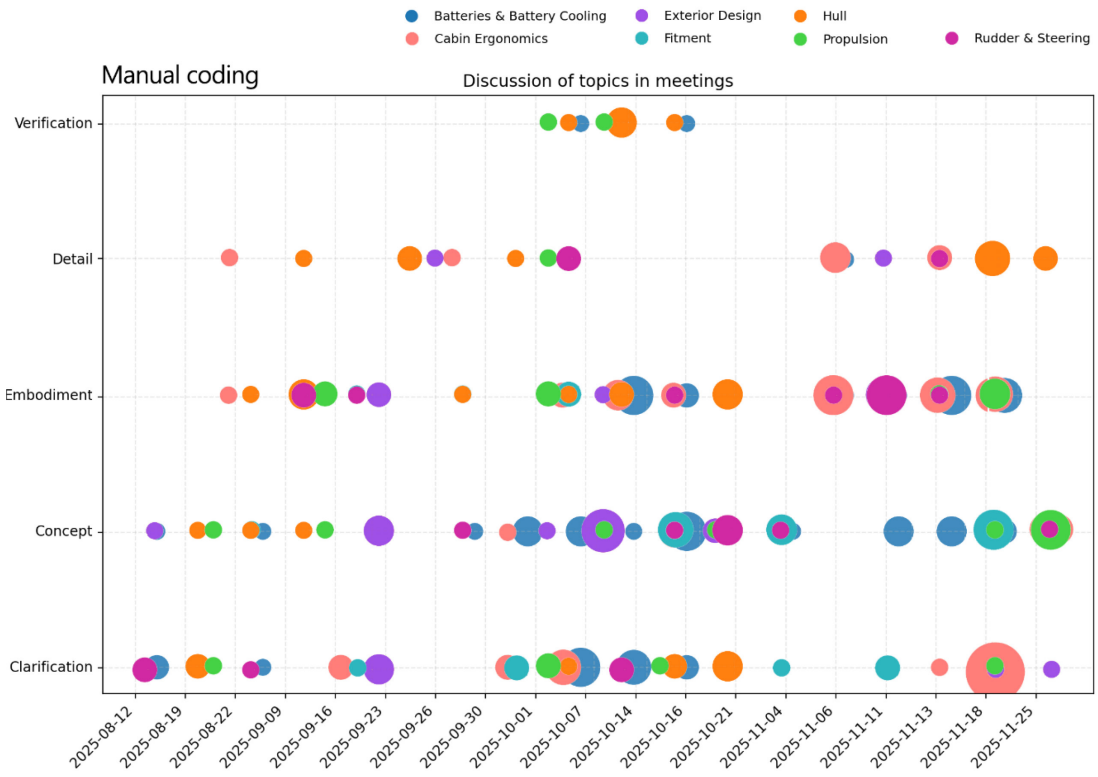


Figure 7. Mapping of manually coded discussion topics.



Figure 8. Mapping of automatically coded discussion topics.

4.2 Expected discussion trends according to VDI 2221

Bubble graph visualizations can be used to examine whether the design process of a given subsystem is following the VDI 2221 guidelines. If VDI 2221 systematic approach is being followed, the discussion relating to the design of a subsystem should begin from the "Clarification" and "Concept" phases on the lower left corner of the graph and then move up towards the "Detail" and "Verification" phases at the top right of the graph (Figure 9).

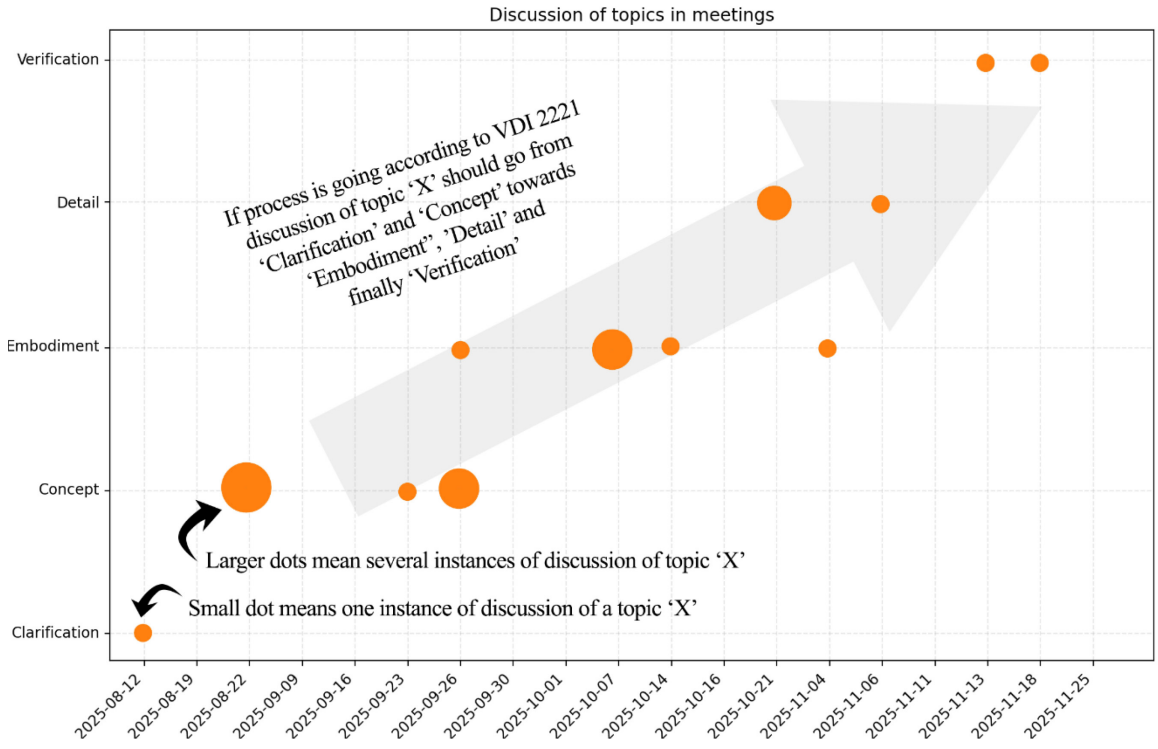


Figure 9. Explanation of bubble graph and expected trend.

4.3 Mapping of individual subsystems

4.3.1 Rudder and steering systems

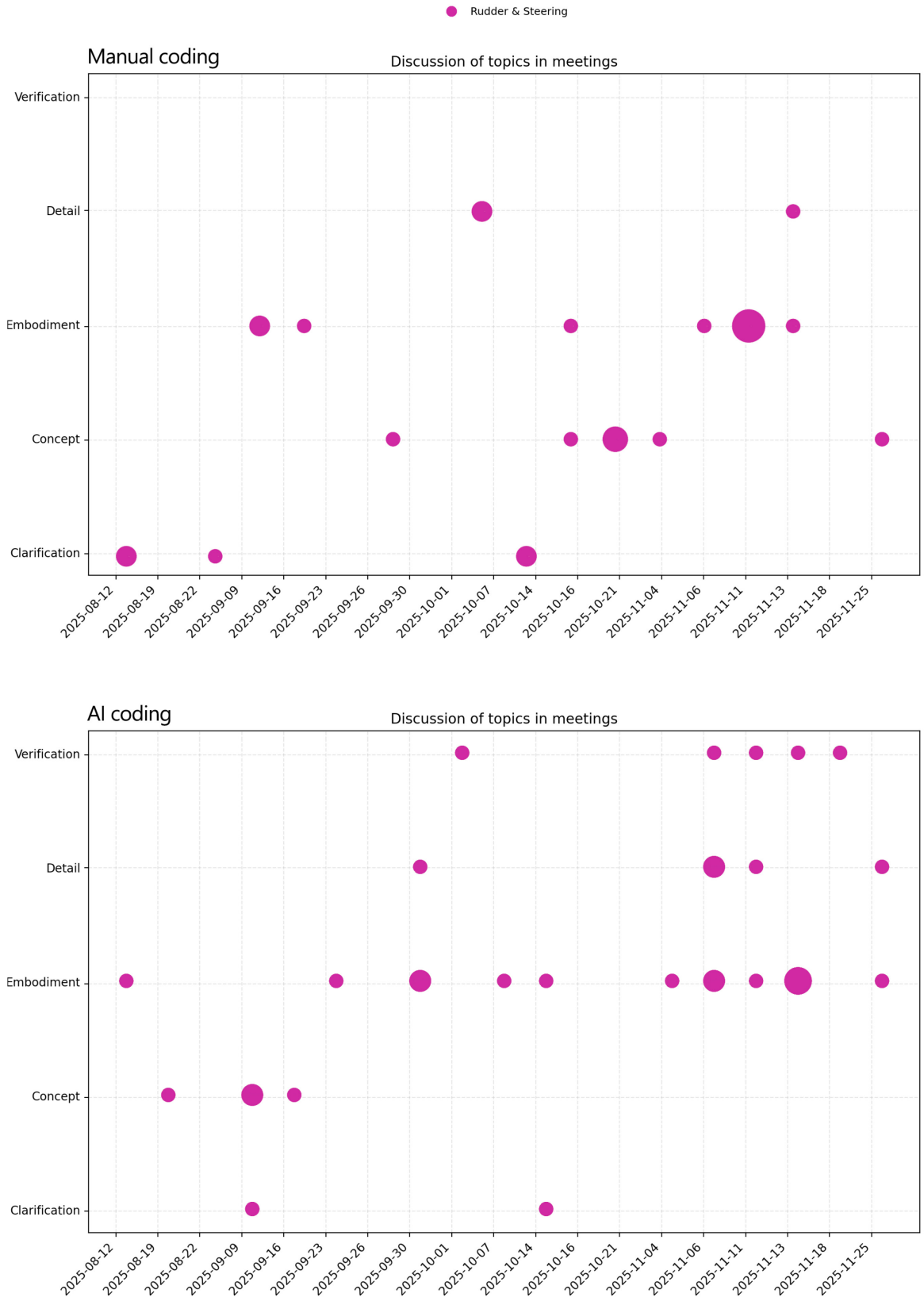


Figure 10. Discussion of rudder & steering topics.

Rudder and steering subsystem showed strong correlation with both adherence to VDI 2221 design phases and between manual and AI coding. And indeed, the practical observations corroborate this result. The Watertaxi project began investigating the rudder design with calculating required steering forces, created alternative concepts for the structure of the rudder and then proceeded to detailed design of the rudder arms and the selection of steering cylinders. An interesting discrepancy between manual and AI coding are the "Verification" items the AI has detected but the human has not. It appears the AI was more diligent in reading the meeting transcripts, and that there had indeed been several items of verification (Table 8).

Table 8. Rudder & steering verification discussion topics observed by AI.

Observation number	Topic	Date
165	Review of galvanic insulator weight and removal of duplicate rudder drive weight.	1.10.2025
364	Verification of steering actuator power requirements by Dometic based on rudder drawings.	11.11.2025
419	Verification of steering actuator power/capacity pending final rudder drawings.	13.11.2025
451	Rudder System Fitment – Checking if the proposed rudder system fits based on the lever arm length provided.	18.11.2025

4.3.2 Cabin ergonomics

Cabin ergonomics -category included all topics that related to how the passengers and the driver were taken into consideration: seats, handrails, roof, stairs, storage spaces, etc. Again, although 1:1 correlation was not evident between manual and AI coding, the trend towards detail design phases is evident. To the embarrassment of the manual coder, the AI was yet again more diligent in recording the details concerning verification (Table 9). However, the human observer, in turn, recorded several important ergonomics-related topics missed entirely by the AI (Table 10), illustrating the danger of drawing premature conclusions.

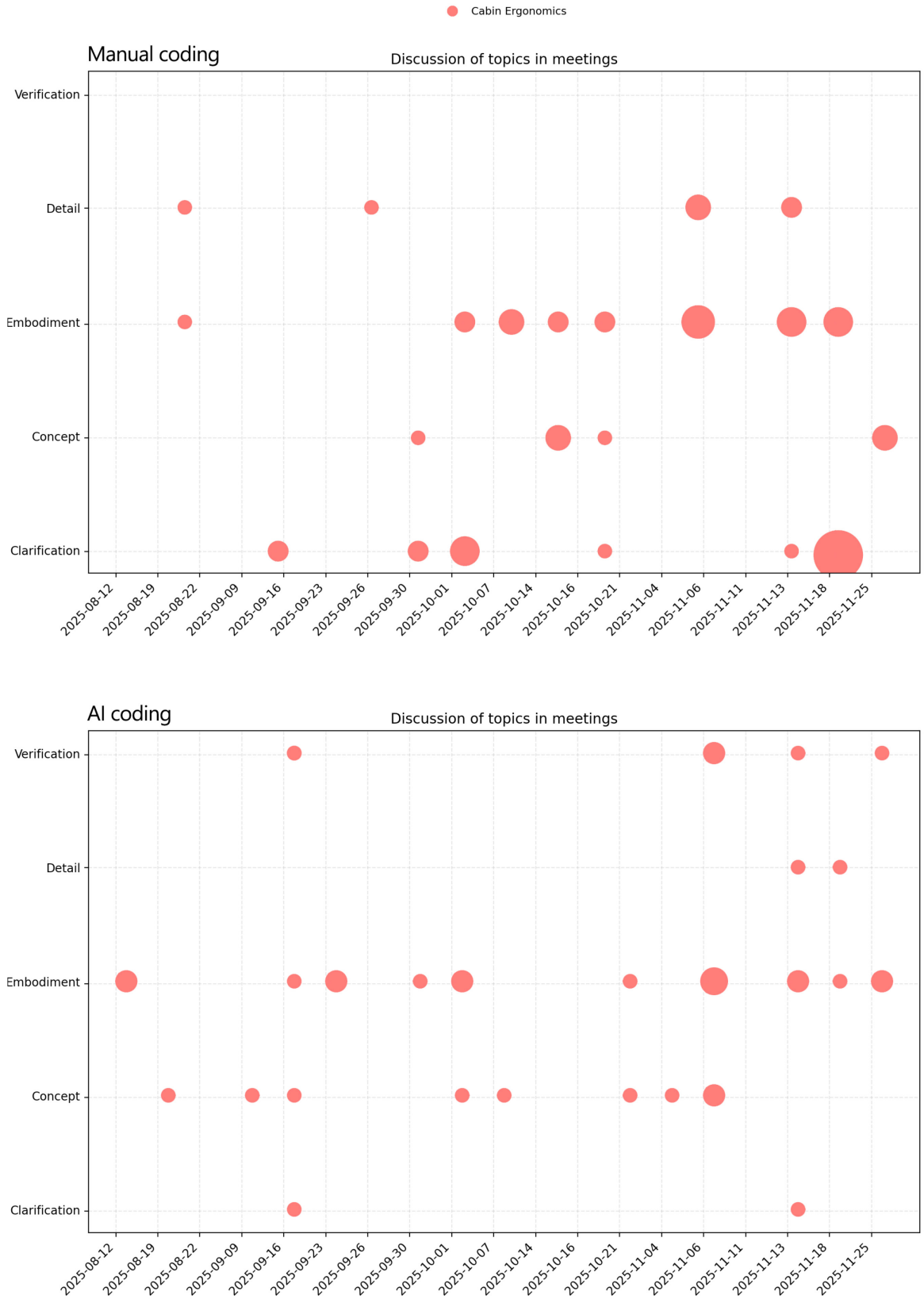


Figure 11. Discussion of cabin ergonomics topics.

Table 9. Cabin ergonomics verification discussion topics observed by AI.

Observation number	Topic	Date
99	Verification of aisle width dimensions (55 cm) against regulations.	16.9.2025
325	Verification of aisle width (600 mm) and door opening (750 mm) against regulations.	6.11.2025
339	Verification of cabin headroom (standing height) to see if the roof can be lowered for bridge clearance.	6.11.2025
372	Verification of door width compliance with harbor regulations (minimum 70 cm).	13.11.2025
500	Driver Seat Height – verifying the seat height relative to the console and visibility.	25.11.2025

Table 10. Cabin ergonomics clarification discussion topics observed manually.

Observation number	Topic	Date
356	Talking about the footrest for the driver and the height of the spring package.	18.11.2025
358	Talking about volume reservation for a more advanced interface for the driver, possibly in the future boats.	18.11.2025
361	Talking about how the driver needs to be able to access the battery charger connector.	18.11.2025
366	Talking about whether the door needs to be closed when the boat is charging during the night. The answer is yes.	18.11.2025
368	Asking can the connector be reached so that it's possible to take it out without coming out of the boat.	18.11.2025

4.3.3 Hull

With the hull design category, a discrepancy between the manual observer and the AI observations emerges. From the point of view of the author, following the meetings in person, the Watertaxi hull design was very systematic, proceeding from rough layout to detailed scantling design. The AI coding, however, differs greatly from this (Figure 12).

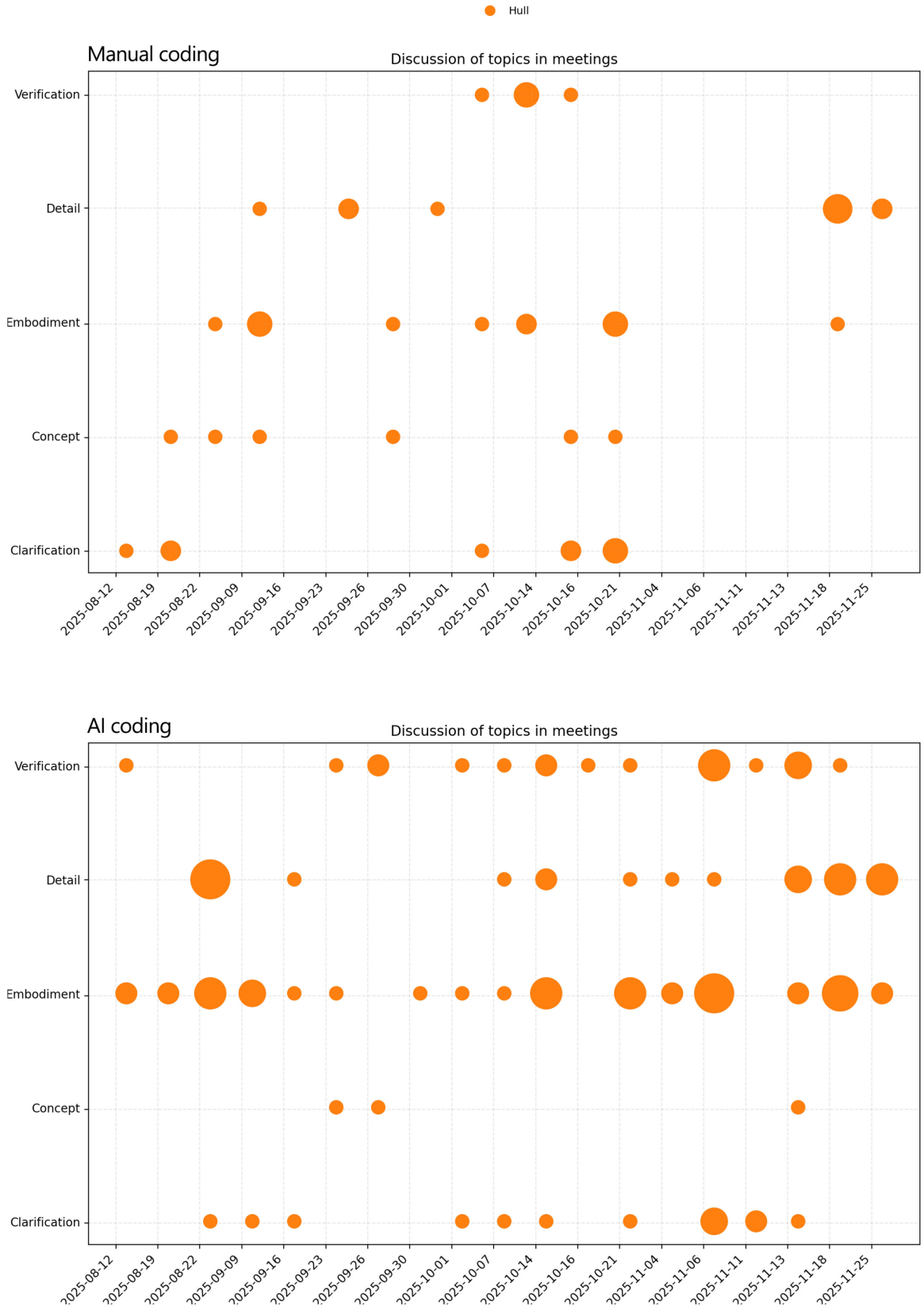


Figure 12. Discussion of hull design topics.

A clue to the discrepancy between the human observer and the AI coding is given by looking at the “Verification” -phase datapoints. Table 11 lists some of the items the AI interpreted to be verification actions. In reality, these observations were not actual completed tasks of verification, merely discussion of what (for example FEA) would be needed to verify a certain detail in the future. There is some indication that the AI is not able to discern between the discussion of a possible future action and an action, which has been completed, in the same way that a human observer can.

Table 11. Verification -stage discussion topics misinterpreted by AI.

Observation number	Topic	Date
131	Discussion on simulation artifacts, specifically the “nose diving” initialization issue in the CFD video.	26.9.2025
132	Verification of wave definitions in the simulation (harmonic vs. cross waves) and amplitude consistency.	26.9.2025
361	Hull outfitting weights: Anodes, cable trays, fender (HDPE estimation), and equipment beds (PMMA).	1.10.2025
193	Damage stability requirement: Ensuring the vessel does not sink if one compartment leaks.	7.10.2025
238	Concerns regarding structural integrity (denting decks, flapping windows) if weight savings are too extreme.	16.10.2025
344	Proposal to perform FEA (Finite Element Analysis) on the aft frame/pump area to reduce structural weight.	6.11.2025
368	Discussion on the necessity of FEA (Finite Element Analysis) to optimize structural weight versus using baseline data.	18.11.2025

4.3.4 Fitment

Fitment refers to the placement of non-structural equipment within the boat. In the context of the Watertaxi, this means items such as inverters, ducting, driveshafts, bilge pumps etc. Although the VDI 2221 stages trend as expected, the manual coding and AI coding differ significantly: manual coding logged 32 data points, while the AI logged 92.

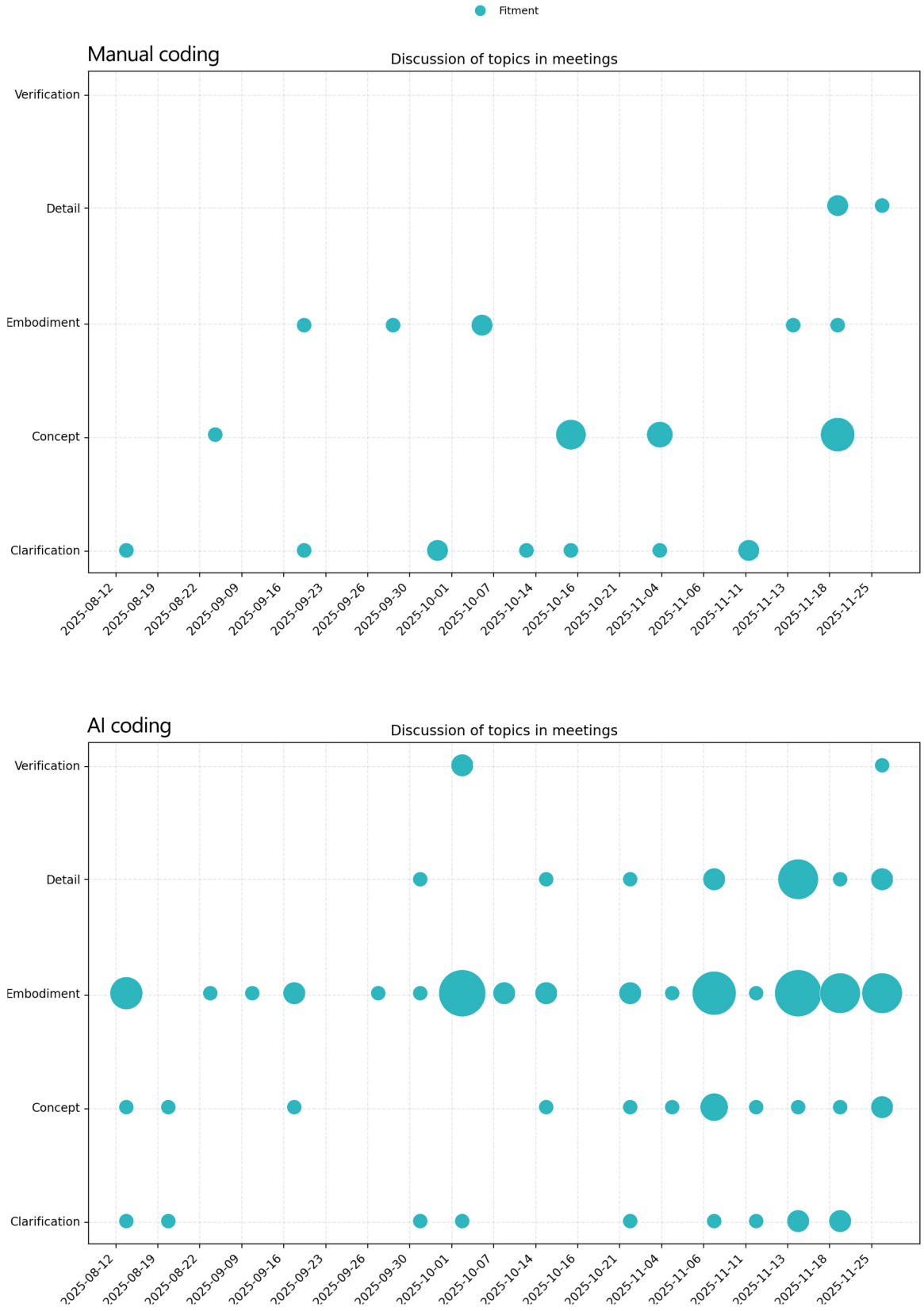


Figure 13. Discussion of fitment topics.

Analysis of the differences reveals that the AI coding is not making mistakes in picking up so many data points. On the contrary, it is able to detect very detailed ideas and allocate them correctly to fitment. Comparing an excerpt of a comments recorded in the meeting transcript (Table 12) with the resulting AI coded observations (Table 13), the level of interpretation the AI is able to do is simply astounding.

Table 12. Excerpt of comments from meeting transcript 13.11.2025.

Time	Comment
1:02:53	*"That was a I I'd imagined it would be there, yeah. So uh yeah. Let's let's assume that. And I think uh we we looked a bit more at the uh high voltage PDU, which is now under the stairs, but we would like to would like to move it to the port side under the foremost bench, and that's that place for the 24 volt batteries. Yep, and we're trying to move as as much as possible from the dashboard uh to port side actually, mostly to make space under the dashboard, and uh also that will probably help in uh leveling the boat in heel."*
1:03:33	*"So okay, good. All right, and then the thing about the 24 volt system, we're now using uh breakers or or how you call it automatic breakers. Um in in this boat, we better use uh just fuses that will uh save a lot of weight and also a lot of uh space. a difference from the one we have now. So I will make the pictures from the boat we have now. Yep. And also mention what's going to change then. All right."*

Table 13. Data coded by AI from input data of table 12.

Observation	Topic	VDI 2221	Subsystem	Date
425	Location of the 24V battery: Moving it from the dashboard to the back or under seats to optimize cabling.	Embodiment	Fitment	13.11.2025
427	Decision to switch from automatic breakers to fuses for the 24V system to save weight and space.	Detail	Fitment	13.11.2025

4.3.5 Propulsion

In propulsion category, manual coding found 26 data points, whereas AI coded 59. In manual coding, delayed decisions on drive shafts and how motors are mounted is visible in the graph.

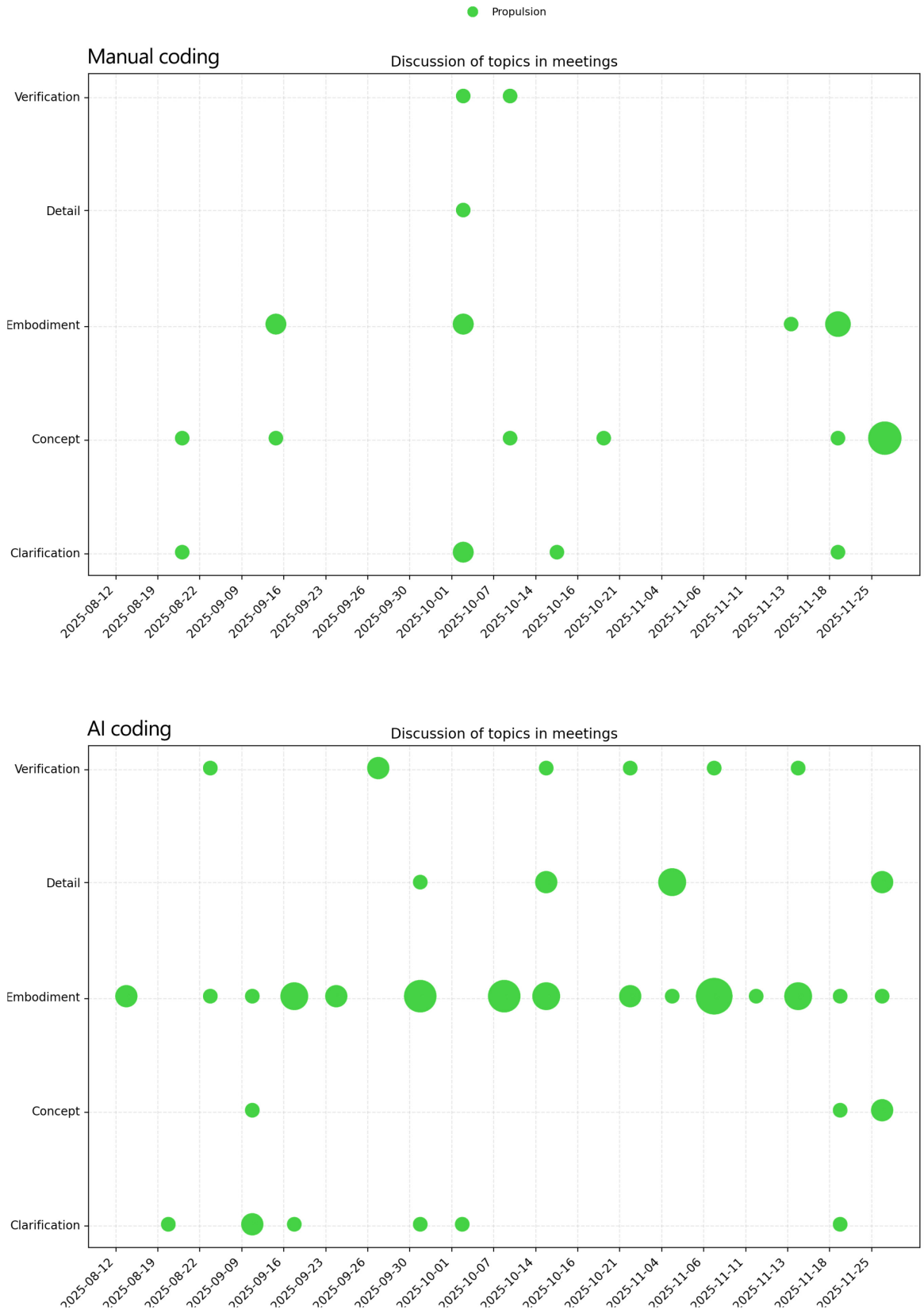


Figure 14. Discussion of propulsion topics.

In propulsion data, attention is drawn to the missing concept phase datapoints in the AI coding. It may be, that discussions related to concepts are often incomplete musings, which are not difficult for a human to notice (Table 14) but the lack of a clear decision point makes them difficult for the AI to pick up.

Table 14. Propulsion topics recorded by manual coding but not detected by AI.

Observation	Topic	Date
62	Talking about bearing: Client prefers bigger bearing due to heavy use; decision made without additional calculations.	23.9.2025
184	Discussion about how inverters could use aluminium directly as heat sinks.	14.10.2025
257	Talking about the pocket area, there's a lot of conflict with the rudder stock and other components.	21.10.2025
374	There's a real problem with how to enter the stern tube through the hull so that it's watertight and can be welded.	18.11.2025
401	Supplier proposal was not good because it had so much wetted area.	25.11.2025
403	Supplier proposal tube needs three bearings on the shaft. The construction is strong, but it's very difficult to integrate into the hull structure.	25.11.2025
405	The engine mounting is still not completely decided whether it's on the front face or somewhere else.	25.11.2025
407	Solving the problem of how to put the shaft into the hull structure happens while in the meeting, moving the frames and the problem gets solved.	25.11.2025

4.3.6 Batteries and battery cooling

The Watertaxi is an electric-powered boat. A substantial amount of the discussions within the meetings were dedicated to batteries and battery cooling. A topic discussed in almost every meeting, it required the project team to return many times to requirement clarification and concept design phases, which is clearly visible in both manual and AI coding.

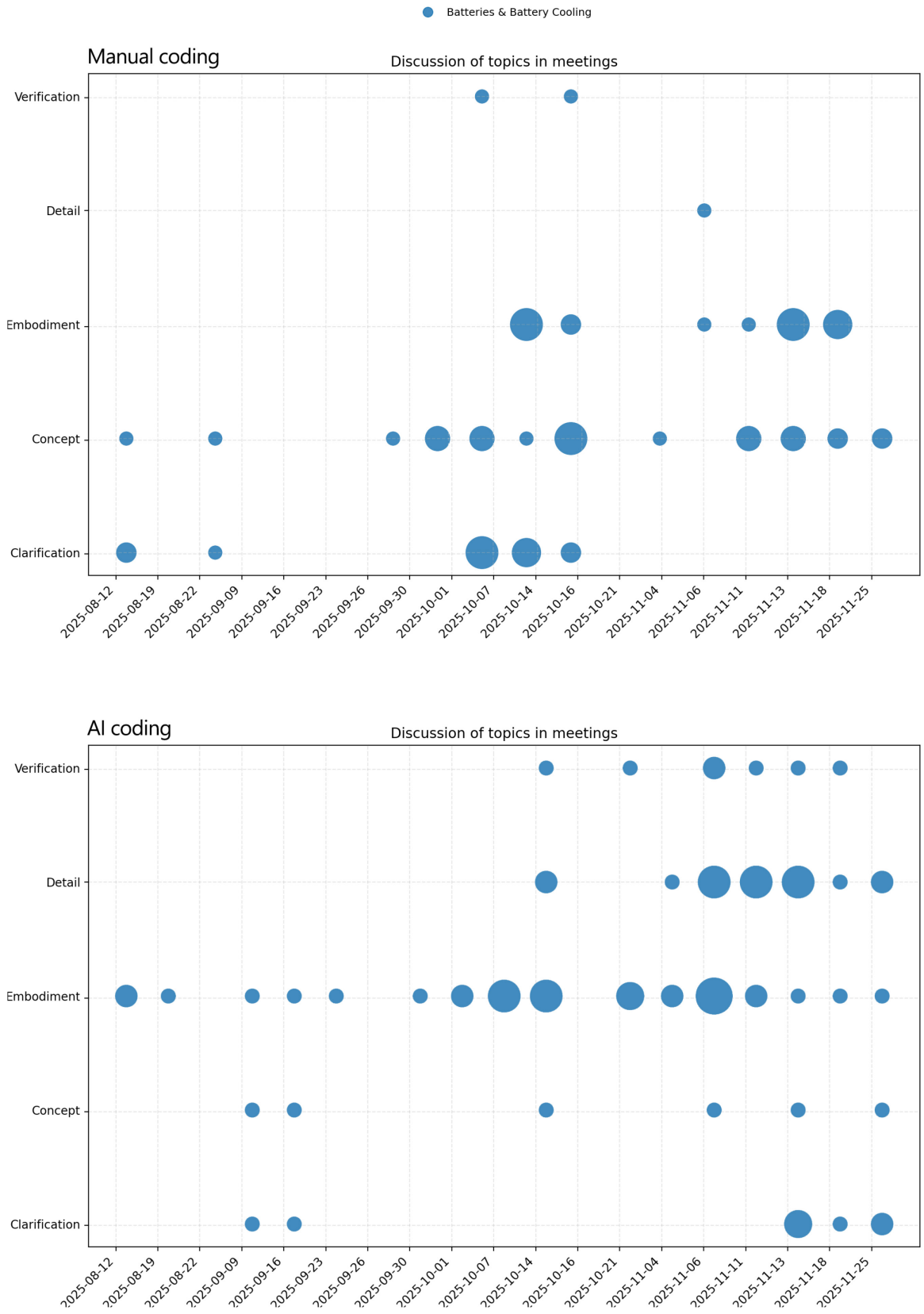


Figure 15. Discussion of battery & battery cooling topics.

4.4 Comprehensive audit of the Watertaxi project

The execution of the Watertaxi project was observed by traditional, manual observation and by automated analysis. When combined, the project meeting logs and project documentation build a high-level summary of the way in which the project was run. The results given here are not an exhaustive list, but a selection of the most prominent observations.

4.4.1 Manual audit of project

The results of manual coding of project flow issues are listed in Table 15. These results are interpretations of problem instances observed by the author during the project meetings.

Table 15. Project flow issues detected by participant observation.

Project flow issue code	Explanation	Number of observations
Communication issue	Practical difficulties in conveying information, for example sharing 3D files, requiring more 2D drawings with dimensions, missing information in component lists, communication issues with suppliers.	70
Requirements unclear	Missing or unclear requirements, for example whether certain regulations apply or whether a feature is needed or not.	63
Concurrency problem	Several tasks should be completed simultaneously, but there simply isn't anyone to do them, as all are occupied with other tasks.	31
Assumptions	An assumption is made to "keep the momentum going." Often related to weight of components, or the ability to reduce weight. Also unfounded estimations of the complexity of designing a system.	31
Mission unclear	Design requires making a decision between conflicting requirements, but the mission target set by the client is not clear.	24
Postponement	Decision is intentionally postponed due to lack of information, for example battery ventilation.	15
Chicken-and-egg dependency	Task A cannot be completed without input from Task B and vice versa. For example, proposal from supplier requires information from designers, designers need a proposal from supplier.	11

4.4.2 Automatic audit of project by AI

The Watertaxi project maintained an extensive "Project Document" of all the important meetings and communication, including excerpts from e-mails. By 25.11.2025 this document extended to 155 pages.

Dr. Marine AI was asked to compare the contents of the project meetings with this "single source of truth" Project Document. The AI was asked to contrast the deductions which could be made from the manually coded observations and the AI-coded observations versus the Project Document and evaluate the adherence of the project to VDI 2221 Systematic Design principles. The following three prompts were used consecutively, and the response from the analysis using Dr. Marine AI is summarized below in Table 16. Full response is included in Appendix 1.

> "Make an analysis of how well this project is following the VDI 2221 systematic design guidelines based on these two observation logs. What are the biggest differences between the two logs with regards to following the VDI 2221 process."
>
> "Compare both first log and second log with this project report document. Which log seems to give a more accurate depiction of the flow of the project?"
>
> "What should be done to improve this boat design delivery process?"

The AI analysis was able to pinpoint "Stage-Jumping", i.e. premature advancement from one design process stage to the next as a significant source of issues. The process change recommendations given by the AI are in line with the general principles advocated by VDI 2221. Of particular interest is the advice given by the AI in point 4, "Integrate Human Factors Early", which was a surprising addition given that it is not a topic mentioned in VDI 2221. This advice may be a direct result of the inclusion of the entire book "Applying Physical Ergonomics to Modern Ship Design" by Olsen (2024) as a RAG knowledge material source (Appendix 4). The key learning here is the importance of providing the AI with not only the core knowledge for a task, but including additional, beneficial supporting knowledge as well in the training material.

Table 16. AI-audit of the Watertaxi project.

Based on the analysis of the project logs, meeting notes, and the systematic engineering principles (VDI 2221) found in your knowledge base, this project suffers from premature advancement. The team is attempting to finalize detailed production files (Stage 4) while still debating basic requirements (Stage 1) and conceptual layouts (Stage 2).	
1. Enforce a Strict "Requirements Freeze" (Fixing Stage 1)	Implement a formal "Change Request" process. If a stakeholder wants to change a requirement (e.g., adding a battery heater), the impact on cost and timeline must be calculated before the engineering team opens the CAD software.
2. Break "Chicken and Egg" Cycles with Interface Definitions	Create "Black Box" models for long-lead items. Design the hull to accept any component that fits within that box, rather than designing around one specific supplier file that might change.
3. Move from "Weight Monitoring" to "Weight Budgeting"	Implement the "Target Weighing" approach. If a subsystem exceeds its allocation, it triggers an immediate "Stop Work" to resolve the variance.
4. Integrate Human Factors Early (Task Analysis)	Require a "Human Interface Walkthrough" using the rough 3D model before detailed engineering begins.
5. Validate Supplier Inputs Before Integration	When a supplier sends a model, it must pass a "Fit & Function" check (Does it meet drag limits? Is it too heavy?) before it is imported into the master assembly.

4.4.3 Interviews with project team members

The final stage of the study was to interview the Watertaxi project team members. The purpose of these interviews was to get one more set of data to compare and validate the findings. From the Agency side, the Watertaxi team involved 3 people. Of the three, one person had moved on to new projects and declined to comment. The two other team members were interviewed 1 month after the observation period. The format of the interview was to first ask questions about the personal observations made by the team members of the project and how it had proceeded. The interviewees had not seen any of the results of the study at this stage but were given an opportunity at the end of the interview to see the results and comment on them. The interview questions are listed in Table 17 and the full answers to the questions are included in Appendices 2 and 3.

The responses to the interviews further validated many of the results given by participant observation and the AI analysis. The answer by interviewee 1 to question 3 is revealing: "The end users of the boat often had not made up their mind on what was actually needed or

wanted, and they only began thinking about their actual requirements once the design had progressed to that part of the vessel". Thinking ahead and changing perspectives to create exhaustive requirement scenarios is a central part of the VDI 2221 methodology.

Comments such as: "Where we failed is that the concept of the boat was supposed to be ready, and then we had to take the concept completely apart during the design." (interviewee 1, question 5), and: "Even if a vessel is based on a previous design, we still need to do the due diligence at the concept phase to avoid the sort of problems we had in this project. In future projects we will ensure we put enough detailed effort into the concept stage." (interviewee 2, question 5), establish that both interviewees regard a successfully concluded concept phase a vital precursor to a well-maintained development project.

On seeing the results obtained by this study, both interviewees agreed, that the results were not surprising: "I am not at all surprised by what's been found here (by the study), those issues were blatantly obvious during the project. The kind of assessment of the maturity of different parts of the concepts is what we would need here also" (interviewee 1). Interviewee 2 commented of the proposed improvements: "This is pretty much what we are working on. AI could be useful in this kind of quality improvement. We are already using AI to record meetings, but I was not aware that it can be enhanced to do such extensive project analysis."

Table 17. Interview questions.

Question 1	What is your working background at the Agency?
Question 2	Was the Watertaxi a typical project?
Question 3	How well did the project progress from your point of view?
Question 4	What kind of distinctive features did this project have?
Question 5	Any learnings from the project you want to mention regarding the work done by the Agency?
Question 6	Are you familiar with VDI 2221 standard?
Comments	Comments on the results presented by the study.

4.5 Experiments with deeper AI analysis

In some respects, making tables of observations and categorizing them using AI, as described in the previous sections, is not harnessing AI to its full potential. The results shown before

in this section involve only 1-2 layers of prompting for interpretation. It was not possible within the scope of this work to go methodically into deeper levels of interpretation of the data. However, it is perhaps important for any further study to demonstrate here, what kind of results were obtained, when Dr Marine AI was asked to extrapolate the interpretation of the analysed data and to draw much more far-reaching conclusions on the data.

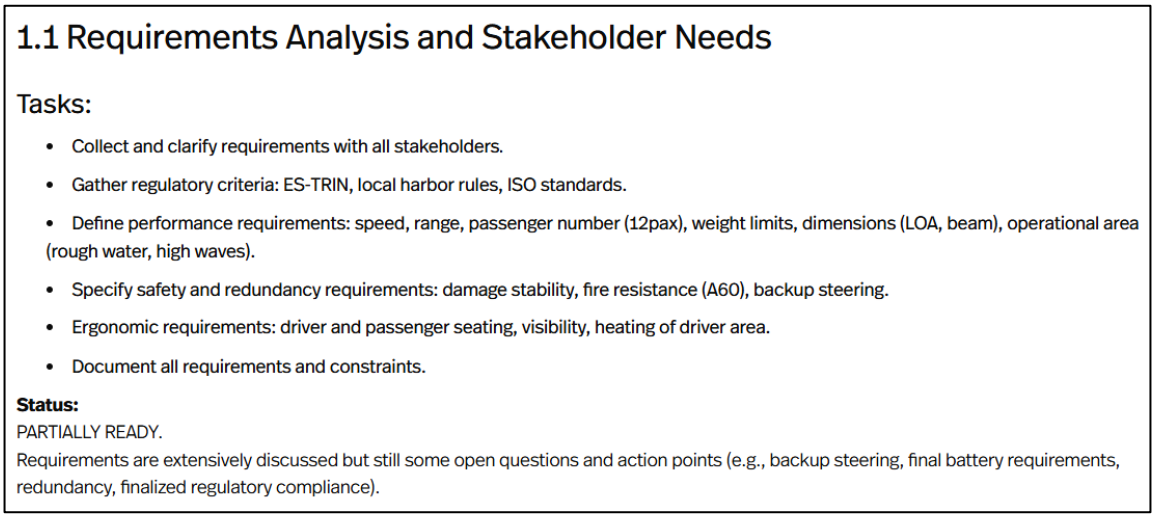
4.5.1 AI-analysis of project progress

Dr Marine AI was asked to evaluate the overall readiness of different ongoing tasks from project document (WT_D1, Table 4) on 13th of November 2025. An abridged response is shown in Table 18 and Figure 16. The prompt given was:

> "Identify design tasks and organize according to systematic design engineering. Try to determine if the design task is ready or unfinished."

Table 18. AI response summary to "identify tasks and their readiness".

Task	Status	Notes
Requirement specification	Partially Ready	Most requirements captured, some open regulatory/ergonomic items.
Concept design (GA, arrangement, main systems)	Partially Ready	Main concepts exist, several subsystems need clarification.
Weight estimation, LCG/VCG	In Progress	Being updated as 3D models and layouts mature
Hull and structure preliminary scantlings	In Progress	Iterative, FEA ongoing, affected by weight and arrangement changes.
Detailed drawings and production files	In Progress	Awaiting concept freeze and finalized arrangements.
BOM and component responsibilities	In Progress	Initial draft exists, needs review and final input.
Equipment fitment and layout	In Progress	3D fitment ongoing, component placements pending.
Battery/fire/ventilation/degassing solutions	In Progress	Proposals made, final solution and supplier integration pending.
Compliance (ES-TRIN, Class)	In Progress	Ongoing, need to confirm requirements and implementation.
Production preparation	Unfinished	Dependent on above, nesting and cut files planned.
Testing and verification	Not started	Will commence after implementation.



1.1 Requirements Analysis and Stakeholder Needs

Tasks:

- Collect and clarify requirements with all stakeholders.
- Gather regulatory criteria: ES-TRIN, local harbor rules, ISO standards.
- Define performance requirements: speed, range, passenger number (12pax), weight limits, dimensions (LOA, beam), operational area (rough water, high waves).
- Specify safety and redundancy requirements: damage stability, fire resistance (A60), backup steering.
- Ergonomic requirements: driver and passenger seating, visibility, heating of driver area.
- Document all requirements and constraints.

Status:
PARTIALLY READY.
Requirements are extensively discussed but still some open questions and action points (e.g., backup steering, final battery requirements, redundancy, finalized regulatory compliance).

Figure 16. AI response detail of project readiness.

4.5.2 AI-analysis of project risks

In addition to knowing the general status of a project, the project manager is interested in having advanced warning to any risks which may be on the horizon. Dr. Marine was asked for a project risk assessment on 17th November 2025 based on document WT_D1. An abridged response is shown in Table 19 and Figure 17. The following prompt was used:

> "What are the highest unresolved risks for the project timeline described in this document."

Table 19. AI response summary to "identify highest unresolved risks".

Risk Area	Impact on Timeline	Status/Notes
Structural Weight Control	Redesign, re-nesting, procurement delays	Not controlled, ongoing concern
Design Freeze/Interface Finalization	Cut file delay, production rework	Incomplete, parts not frozen
Supplier Lead Times/Procurement	Missed build milestones, cascading delays	Long lead items not ordered
Regulatory/Compliance Uncertainties	Late redesign, inspection rework	ES-TRIN details still evolving
Component Integration/Fitment	Assembly conflicts, last-minute rework	3D fitment and space still open
Outstanding FEA/Engineering Analyses	Redesign, schedule slip for structural works	Several FEA studies not complete
Design Change & Communication Management	Rework, confusion, missed dependencies	High volume of changes, poor freeze

Most Critical Unresolved Risks

1. Lack of firm weight control (structural, outfitting, and systems) 2. Delayed design freeze and incomplete interface definition 3. Pending regulatory interpretations and supplier lead times 4. Outstanding engineering analyses (FEA and fitment)

Each of these can cause cascading project delays if not resolved promptly.

Mitigation recommendations:

- Immediate design freeze for hull, main structure, and battery systems.
- Complete all outstanding FEA and integrate results before cut files.
- Prioritize procurement of long-lead items based on “frozen” specifications.
- Assign a dedicated compliance manager to track ES-TRIN and local authority requirements.
- Intensify cross-discipline coordination meetings until all main interfaces are closed.

Figure 17. AI response detail when asked for project risks.

5 Discussion

5.1 Systematic Design principles versus reality

The results demonstrate, that although the Watertaxi project employed a productive concurrent engineering workflow, the process appeared at times to be at odds with the principles of the VDI 2221 systematic design guidelines.

5.1.1 Stage-jumping

The VDI 2221 guidelines advocate a step-by-step progression from clarification of initial requirements to concept design, embodiment design and then to detail design. The Watertaxi project exhibited frequent “stage-jumping”. This was not because the design team was unaware of the dangers of leaping into detailed design too early. Stage-jumping was an unavoidable result of overly tight timelines and “chicken-and-egg” dependencies, where something had to be proposed, even in the absence of clearly stated requirements, to move the project forward.

5.1.2 The hidden cost of agility

Although fast response is a virtue in some instances, the agile implementation led to rework and strain on the project team. Skipping proper requirements definition expected in VDI 2221 Stage 1 increases the risk of substantial latent problems in the design and postponing clarification often only creates “technical debt” – debt, which must, in any case, be paid as extra work during the embodiment and detail phases.

5.2 AI as a project management advisor

A primary goal of this thesis was to determine if AI can indeed benchmark the execution of a complex project by analysis of the project meeting logs and documentation. In many instances, the AI proved to be a more proficient observer than a human. Unlike a human

observer, the AI does not get tired of focusing on hundreds of pages of meeting transcripts. One 1-hour long meeting took 45 minutes to code manually and yielded 32 observations. The same meeting, when processed by Dr Marine AI, took 15 seconds to analyze and log observations. In categories like Fitment, the AI detected 92 data points where the human only noted 32, showcasing that the AI is meticulous in sifting through the material it is given. In terms of quantity and speed of work that can be done by AI, there is no question that it can be a tool of almost real-time analysis of a meeting or the overall situation of a project.

5.2.1 Shortcomings of AI in context and implicit reasoning

Although prodigious in output, the reasoning of even a RAG-enhanced GenAI was not on par with the human observer. Firstly, the AI struggled to understand the difference between events which had taken place and those planned to be done in the future. The AI routinely flagged a verification stage completed when the project team was in fact discussing what verification would be needed to be done in the future. A human observer was also superior in recognizing concept design -phase discussions, which often lacked a clear direction from a start to an endpoint, being more of exchanges of ideas than decisions. In conclusion, the reasoning of implied intent behind a conversation remains something where a human observer is clearly more adept than the AI.

5.2.2 Improved reasoning by a RAG-enhanced GenAI

The use of Retrieval-Augmented Generation (RAG) clearly improved the context-specific reasoning of the LLM. Giving the AI agent additional specific knowledge of naval architecture and VDI 2221 systematic design process improved its interpretation of the project meetings and documents. For example, the RAG-enhanced AI avoided the basic mistakes made by the general-purpose Sembly AI's, which, for example, labelled all instances of "scantling" as "scanning".

5.2.3 Leveraging AI analysis even further

In the course of this study, it became apparent through experimentation (4.5) that much more could be done with automated project analysis. This research is only the beginning of what could be possible. By extending the AI analysis of the project tracking to multiple-level interpretation of the results, automation could be made to give timely warnings of budget overruns, gradually developing engineering conflicts and other very high-level, even abstract project management issues. The technology to do so exists. Making it usable and reliable requires much more study.

5.3 Recommendations for managing a marine design project

Based on the audit, several process changes are recommended for the Agency.

5.3.1 Going beyond the "Ship Design Spiral"

A methodology of designing marine vessels, the "Ship Design Spiral", was proposed by professor J.H. Evans in his foundational article, "Basic Design Concepts", published in the American Society of Naval Engineers Journal, November 1959 (Figure 16). The underlying principle and motivation of Evans' spiral is that ship design, especially prior to computer-aided design, was a laborious iterative process. Evans' idea was to bring systematic procedure to the sequential calculations and considerations and thus make convergence on a single solution less onerous. Even though Evans' Ship Design Spiral has been superseded in many cases by more modern methodology, it is still used in naval architecture. Evans' influence was visible in the course taken by the Watertaxi project.

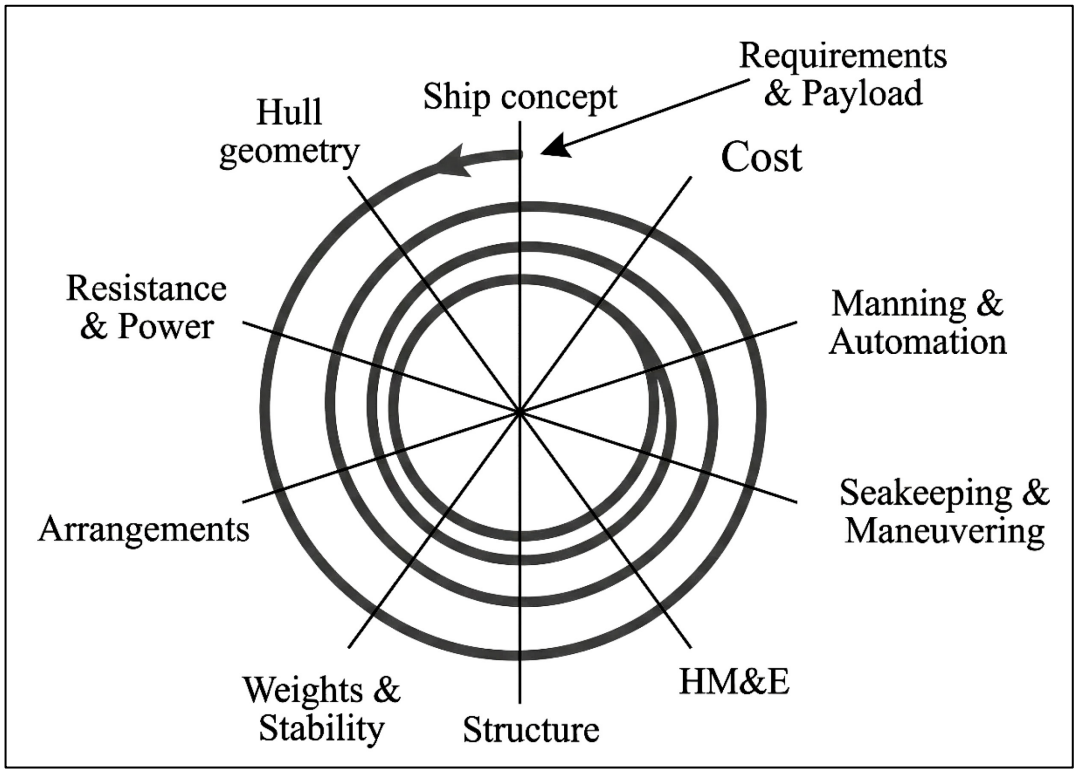


Figure 18. Evans' Ship Design Spiral (based on Karczewski & Kozak 2017, 1).

The challenge of the Ship Design Spiral is that it is a traditional "Point-Based Design" (PBD) philosophy. In PBD, the design iteration is intended to proceed from one firm point of intermediate results to another, and finally to converge on a working final concept. The problem this approach is, that the spiral will fail to converge, if the technological maturity of the different "rungs" of the ship design spiral are completely out of step with each other. For example, in the case of the Watertaxi, the propulsion subsystem concept was far from ready at the onset. Adapting to the "technical debt" of the propulsion subsystem hindered progress of other components of the boat.

The philosophy of VDI 2221 could be used to mitigate these issues. VDI 2221 systematic design process is an example of "Set-Based Design" (SBD). SBD is widely used in product and automotive design. The idea behind SBD is that instead of trying to maintain and improve a monolithic concept of a new product as in PBD, the design problem is split into sets of problems. Each of the problem sets is tackled separately. The final concept is a solution which lies at the intersection of these sets. A famous example of SBD design

philosophy was the Wright Flyer. The Wright brothers won the race to powered flight by dividing the “obstacles of flight” into three subsets: in-flight control, wing profile optimization and propulsion, and then solving each subset independently, e.g. by building one of the world’s first wind tunnels. The successful Wright Flyer was the first monolithic prototype, where all three subsystem solutions were combined, and a prime example of how SBD approach can be used to ensure a successful design outcome.

5.3.2 Using VDI 2221 in project planning and preliminary concepting

In the case of the Watertaxi project, many of the problems were inherited from the insufficient project planning phase. An approach which could be used to create a more realistic appraisal of the baseline situation before the project, would be to use VDI 2221 stages as a metric of “maturity”. Each subsystem would be evaluated separately. For example, the project planners could ask “is the propulsion system for this boat in clarification, concept or detail maturity phase? How does it compare to what we know of the hull, steering and other subsystems?” If the appraisal reveals, for example, that a particular subsystem is completely unresearched and at early clarification stage, and others are at detail design (for example, can be copied over from an existing previous project), then project planners should first attempt to “bridge the knowledge gaps” before committing to a fixed solution to be delivered at a fixed time.

The proposal here is not to replace the tried-and-tested Ship Design Spiral with the VDI 2221 approach or set-based design. The proposal is to use the methodology of VDI 2221 to create a better high-level comparative analysis of the maturity of subsystems, bridge substantial knowledge gaps before attempting to proceed along the traditional Ship Design Spiral.

5.3.3 Weight budgeting

The Watertaxi project used weight tracking and had some success in weight reduction. However, often a surprise increase in weight was simply acknowledged and accepted. Pro-active “weight budgeting” is a different approach. The whole boat and subsystems are

allocated weight budgets based on performance expectations. These budgets are “used up” by components. Exceeding a budget limit triggers a “stop work” event, where either the performance target and budget is adjusted, or additional work is commissioned to reduce the weight spent on a certain solution. This makes weight targeting the shared problem between the client and the design agency, instead of becoming an impossible mission for the designers.

5.3.4 Early Human Factors integration

In the Watertaxi project, many ergonomic details became visible only weeks after the project had begun. Human Factors is a distinct problem area, and it is often one of the first subset of requirements which is mapped in product and automotive design projects. There is no reason not to use the same approach in a boat design project. Systematic focus on Human Factors as a separate topic generally improves usability, safety and client acceptance of designs.

5.4 Revisiting the research questions

Question 1: “Can artificial intelligence be used to analyze and assist in the complicated process of marine vessel design?”

Answer 1: AI can be used for that purpose. A suitably retrieval-augmented GenAI is able to map with surprising accuracy the course of design decisions within a longer time span of the project. The advantage of an AI is that it can almost instantly recall and summarize details from hundreds of pages of project team interaction. Doing a similar review of the proceedings “by hand” (as was done in this study) requires a time-consuming process of revisiting meeting notes and project documents

Question 2: “What specific patterns of non-adherence to VDI 2221, if any, does the AI detect in a real-world naval engineering project?”

Answer 2: AI-powered examination of the real-world project logs using VDI 2221 guidelines as a “yardstick” of assessment highlighted “stage-jumping” as a recurring issue. This observation was corroborated by both manual participant observation and further

corroborated by the interviews. However, it must be noted that this stage-jumping was in most cases a conscious choice forced on the project team by the tight project schedule.

Question 3: "How do these AI-detected deviations correlate with results from a traditional participant observation done by a human?"

Answer 3: In many cases, the AI-detected deviations were in line with those picked up by manual participant observation. The most notable differences in the AI-powered and manual assessment were in the interpretation of "Verification" and "Concept" stages. The AI often interpreted verification actions as done, when they were actually only being planned. In discussions of conceptual design, AI had trouble detecting the outcome of the open-ended discussions. On the other hand, for a human observer going over hundreds of pages of meeting transcripts is tiring, and the observer begins to miss details. AI automation was tireless in this regard, and recorded details that were missed by the human observer.

Question 4: "What is the potential of AI tools to assist in real-time monitoring of the trajectory of a project?"

Answer 4: The potential of AI for real-time monitoring of an engineering project has not been tapped at all to the extent that it could be. Simple meeting transcripts and summaries are only the first steps of what could be done. If the engineering process was recorded in a way that is accessible to AI analysis and a systematic contextual framework (such as VDI 2221) was used, the AI analysis would no doubt be able to give useful feedback on where the project was succeeding and failing and possibly also to suggest a course of action.

6 Conclusions

The study found that a suitably trained, RAG-enhanced AI can create a comprehensive "bird's-eye view" of a project in a fraction of the time required by a human analyst. By using VDI 2221 as rubric of assessment, the AI successfully identified systemic issues such as premature advancement between stages. These results were largely corroborated by direct human observation and by interviews with project team members.

AI proved to be diligent in data extraction, in many cases identifying more data points from meeting transcripts than a human observer. The AI-coded data points were examined by comparing them closely with meeting recordings. More often than not, the RAG-enhanced AI had correctly identified and labelled a discussion of a particular topic. While the AI was tireless in its focus, in e.g. catching more details of fitment decisions, it was less able than a human observer to correctly assess VDI 2221 design maturity levels. It was also poor in making the distinction whether a decision had been made or was planned to be made.

The timing of this thesis work was fortunate: at the time of writing, new applications of artificial intelligence are erupting into the market. This study was an experiment of using AI to analyse human processes which are difficult to quantify and visualize. AI tools clearly have untapped potential for project tracking and management. Instead of merely recording meetings, a suitable AI could provide real-time feedback on the progress of the project and suggest actions (4.5). In order to do so effectively, the project meetings and workflow should follow a systematic approach and use well-defined concepts such as the design phases described by VDI 2221. This would help the context awareness of the AI and improve the quality of the analysis. Time-tested traditional project management frameworks, such as the VDI 2221 standard, could have a new purpose in the age of artificial intelligence.

This study suggests, that much more could be done with AI in the context of project management. AI could give real-time advice, alert teams to emerging engineering conflicts and develop strategies. The real limitation, it seems, is not the capabilities of AI as a tool, but our imagination on how it should be used - an enticing topic open for further experimentation!

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Appendix 1. AI-automated audit of the Watertaxi project

Prompts given to Dr Marine AI agent:

> "Make an analysis of how well this project is following the VDI 2221 systematic design guidelines based on these two observation logs. What are the biggest differences between the two logs with regards to following the VDI 2221 process."
>
> "Compare both first log and second log with this project report document. which log seems to give a more accurate depiction of the flow of the project?"
>
> "What should be done to improve this boat design delivery process?"

Unabridged response:

Based on the analysis of the project logs, meeting notes, and the systematic engineering principles (VDI 2221) found in your knowledge base, this project suffers from premature advancement. The team is attempting to finalize detailed production files (Stage 4) while still debating basic requirements (Stage 1) and conceptual layouts (Stage 2).
1. Enforce a Strict "Requirements Freeze" (Fixing Stage 1)
The logs show that critical constraints (battery heating, rudder type, harbor zones) were still being debated in November, months after the project start. **Action:** Halt detailed 3D modeling of non-critical areas until a signed Requirements Specification is locked. **Why:** You cannot optimize a hull structure if the buoyancy requirements (Harbor Zone 2) or heating systems (glycol vs. electric) change weekly. **Process Change:** Implement a formal "Change Request" process. If a stakeholder wants to change a requirement (e.g., adding a battery heater), the impact on cost and timeline must be calculated before the engineering team opens the CAD software.
2. Break "Chicken and Egg" Cycles with Interface Definitions
The project stalled repeatedly because the team waited for supplier 3D models (e.g., driveshaft, batteries) before designing the surrounding structure. **Action:** Define Abstract Interfaces first. Instead of waiting for a supplier's STEP file, define the volume, weight limit, connection points, and max drag allowed for the rudder. **Why:** The logs show the team received a rudder model that was "too heavy" and caused "excessive drag." If they had sent performance specifications to the supplier first (rather than just asking for a model), this iteration loop would have been avoided.

Process Change: Create “Black Box” models for long-lead items. Design the hull to accept any component that fits within that box, rather than designing around one specific supplier file that might change.

3. Move from “Weight Monitoring” to “Weight Budgeting”

The logs describe weight estimation as a passive activity that resulted in an “alarming” 380 kg surplus and a critical LCG shift.

Action: Assign Strict Weight Budgets to every subsystem (Hull, Fitment, Propulsion).
Why: Currently, the team “estimates” weight as they design. Instead, the Hull team should be told: “You have 2100 kg. If you exceed it, you must redesign or request a budget increase from the Project Manager.”
Process Change: Implement the “Target Weighing” approach. If a subsystem (e.g., interior joinery) exceeds its allocation, it triggers an immediate “Stop Work” to resolve the variance, rather than waiting for a full-vessel calculation at the end of the month.

4. Integrate Human Factors Early (Task Analysis)

The logs reveal that driver ergonomics (charging port access, mooring reach) were treated as afterthoughts, leading to late changes in the cabin structure.

Action: Perform a Task Analysis (as per the Application of Human Elements text in your knowledge base) for critical operations: Mooring, Charging, and Emergency Egress.
Why: The design team realized late that the charging port location made it hard for the driver to reach. A simple storyboard of “The Charging Process” in the Concept Phase would have highlighted this constraint before metal was cut.
Process Change: Require a “Human Interface Walkthrough” using the rough 3D model before detailed structural engineering begins.

5. Validate Supplier Inputs Before Integration

The project suffered because supplier inputs (like the stern tube) were integrated into the design before being validated against the requirements.

Action: Implement a Supplier Data Review Gate.
Why: The team integrated a stern tube model that was hydrodynamically inefficient (17-27% added resistance), forcing a redesign.
Process Change: When a supplier sends a model, it must pass a “Fit & Function” check (Does it meet drag limits? Is it too heavy?) before it is imported into the master assembly.

Appendix 2. Interview with Watertaxi project team member 1

Question 1: **What is your working background at the Agency?**
"I have only recently began working as a full-time engineer at the Agency. However, I have been a part-time employee at the agency for several years. I have been completing my studies in naval engineering at the same time."
Question 2: **Was the Watertaxi a typical project?**
"The Watertaxi was not a typical project at the Agency, but it was a typical project with this Client. I was a member of the design team which did the previous design for the same client, and their way of working was familiar to me."
Question 3: **How well did the project progress from your point of view?**
"The project has now used two months more for the design than what was scheduled. Despite it we are still under budget for amount of design hours. We should have had more people on the team, and more of the work we have done should have been done already at concept stage. The project itself was a very similar project to the previous project for the same client. On the other hand, the project was managed to a great detail from the Client side, but on the other hand critical information emerged far too late and the hull needed redesign at very late stages. The end users of the boat often had not made up their mind on what was actually needed or wanted, and they only began thinking about their actual requirements once the design had progressed to that part of the vessel."
Question 4: **What kind of distinctive features did this project have?**
"The distinctive feature of this project versus our typical projects was the that the end users (Client's client) were struggling to communicate their requirements. They had decades of experience of operating these types of vessels, and once they understood what we were asking, they were able to go into a lot of detail on how something should work on the boat. The problem was that they were assuming that we were somehow aware of something that had not been expressed in the project requirements or project meetings. As a result, lots of things that were self-evident to them only became apparent to us, the designers, often very late."
Question 5: **Any learnings from the project you want to mention regarding the work done by you and the Agency?**
"The fact that we were able to create a first draft of the hull in just 1,5 weeks for the purpose of weight estimation was an achievement. The core engineering effort was very successful in that respect. Where we failed is that the concept of the boat was supposed to be ready, and then we had to take the concept completely apart during the design. This was an unusual situation for us. The Client was also pushing us to release only partially ready designs of details or partially researched information (such as weight) but which then was taken as facts, leading to unpleasant surprises later.

However, have already started a new project with the Client and these problems have been addressed and taken into account from the beginning this time.”
Question 6: **Are you familiar with VDI 2221 standard?**
Not familiar.
Comments on the results presented by the study
“I am not at all surprised by what’s been found here (by the study), those issues were blatantly obvious during the project. The kind of assessment of the maturity of different parts of the concepts is what we would need here also.”

Appendix 3. Interview with Watertaxi project team member 2

Question 1: **What is your working background at the Agency?**
"I have been at the Agency 3,5 years now. I have a lot of experience in steel and aluminium engineering from various companies. I studied naval engineering. At the Agency I lead projects and quality improvement."
Question 2: **Was the Watertaxi a typical project?**
"The Watertaxi was a typical project for this Client. Big changes to design at the end stretch of the project. In comparison to other clients, the concept was not as frozen as it usually is and as a result, fundamental things changed during the detailed engineering phase."
Question 3: **How well did the project progress from your point of view?**
"The project was a bit all over the place."
Question 4: **What kind of distinctive features did this project have?**
"This client has an R&D mindset – they want to develop things all the time as they go along. This is an unusual setup for us. Normally clients come to us with a clear idea of the boat they want, and we finalize and execute the design on that idea. That was not the case here. In fact, in a new project with this Client we have already agreed on a different approach which is more suited to the kind changes they may require."
Question 5: **Any learnings from the project you want to mention regarding the work done by you and the Agency?**
"We learned that we need to be more involved in the concept stage of the new designs. Even if a vessel is based on a previous design, we still need to do the due diligence at the concept phase to avoid the sort of problems we had in this project. In future projects we will ensure we put enough detailed effort into the concept stage."
Question 6: **Are you familiar with VDI 2221 standard?**
Not familiar with the name, familiar with the systematic design idea.
Comments on the results presented by the study
"This is pretty much what we are working on. AI could be useful in this kind of quality improvement. We are already using AI to record meetings, but I was not aware that it can be enhanced to do such extensive project analysis."

Appendix 4. Non-parametric text and data mining sources used in Dr. Marine AI

Source	Use
Pahl, G. et al. P. 2007. Engineering Design: A Systematic Approach (Purchased from Springerlink)	Text and Data relating to the implementation of the VDI 2221 process.
EMSA Battery Guidance v1.0 (freely available online)	European Maritime Safety Association Guidance on the Safety of Battery Energy Storage Systems.
Olsen, A.A. (2024). Applying Physical Ergonomics to Modern Ship Design (Purchased from Springerlink)	Regulations ang guidelines for ergonomic design on marine vessels.
Bertram, V. (2012) Practical Ship Hydrodynamics. (Purchased from Elsevier)	Hydrodynamic guidelines for hull and propulsion design.
ES_TRIN 2025 (freely available online)	European Standard laying down Technical Requirements for Inland Navigation vessels (ES-TRIN) from European Committee for drawing up Standards in the field of Inland Navigation (CESNI).